

The S2 pilot measurements — campaign report (single execution of 2026-10-04)

Draft — not published. English translation of the accepted report; numbers, verdicts and sealed labels are identical to the original; the sealed package and the original outputs prevail in case of discrepancy. Report only: supporting documents are provided on request, under agreement.

Translation conventions. Printed outputs, verdicts and sealed labels are kept in French, unchanged: in code blocks, between French guillemets, or as bare labels in the running text. At its first occurrence in the running text, each such passage carries an English gloss, in italics between square brackets, unless it is already legible in English (symbols, English terms); code blocks are not glossed; the glossary at the end of this document lists the glosses. Quotations from other French-language documents of the project (design document, ADR and annexes, sealed package, project journal, recorded answers of the investor) are given in English translation between English quotation marks; the French original prevails. Numbers follow English usage (decimal point, comma as thousands separator), except in identifiers, hashes, ISO dates and copied outputs.

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Status: report drafted on 2026-10-04 (UTC) by a fresh drafter (model `claude-opus-5-5`), who wrote neither the analysis code, nor the pre-registration package, nor the renderings. It reports what the renderings of the single execution print, without reinterpreting the rule: the sealed text is authoritative (package §10.2, pts 1 to 11; pt 11: an unforeseen case is a declared deviation, never a rewrite).

Basis: ADR-0028 D6 (v) (output wired to `docs/11-mesures-pilotes.md`), D6 (vi) and D9; exit criterion of `docs/10-mesures-pilotes-design.md §7`; sealed package `docs/adr-0028/PAQUET-PREREG-S2.md` (sha256 `4d2a8276316b1c66a08aabb15ff3b39be812978e93dafc01a4e627f20d0af528`); renderings filed byte for byte in `docs/adr-0028/execution/rendu-2026-10-04/`.

Provenance: G1 journal of the drafter; it lists each reading, each command and its output. Each number in this report is either copied from a rendering (file and line cited) or computed by a script of the drafter (mark `[calc]`, output cited). A computation by the drafter never replaces a printed value.

1. Summary

Reading conventions and lexicon: next section.

Question. S2 was to settle the question posed by the roadmap: "are the R2 axes observable in practice, and does the R1 test discriminate anything on real data?" (doc 10 §1). The exit criterion sets the deliverable: "n windows, K, z of the pool, the observed R2 partition, k_{eff} vs nominal k; negative result = result" (doc 10 §7). The decision rule is the sealed rule SHOGEN-CRITERE-R1-1 (package §10.2), evaluated only once on the J28 segment.

Verdict of the rule. On J28 (35,982 one-minute windows after the D5 exclusion; analysis pool of 11 feeds and 10 hosts after the removal of Pyth), the rule returns, in each of the two strata, the value **NE REJETTE PAS [does not reject], on account of the discordance**: the confirmatory statistic z_s exceeds 2.33 (≈ 20.8 in calm, ≈ 28.5 in stress), but the block standard-error floor brings it back below the threshold ($z_{\text{bloc},s} \approx 1.95$ and ≈ 1.74). « **R1 discrimine** » = **FAUX** [*"R1 discriminates" = false*] (J28 I.435522). For each stratum, the rendering prints the sealed statement of the discordance case: « le modèle binomial de doc 10 §5.1, à fenêtres indépendantes, est rejeté ; la cause n'est pas identifiée entre co-défaillance des sources et dépendance sérielle des fenêtres » [*the binomial model of doc 10 §5.1, with independent windows, is rejected; the cause is not identified between co-failure of the sources and serial dependence of the windows*] (J28 I.435517, I.435520), with a minimum detectable excess $\text{EMD}_s \approx 196$ windows in calm and ≈ 211 in stress. On the R2 side, the ASN axis was observed: the observed partition counts $k_{\text{eff}} = 4$ classes for nominal $k = 10$ hosts (J28 I.435779-435780); flag 1 is false in both strata and flag 2 is off (J28 I.435794-435795).

What this verdict allows one to say. On 24,585 calm windows and 11,397 weekend windows, outage, staleness and out-of-envelope axes, as observed by this instrument (host, DNS and network of the harness included): the binomial model with independent windows is rejected in both strata (sealed statement of the discordance); the pre-registered rule, which also requires $z_{\text{bloc},s} \geq 2.33$, returns NE REJETTE PAS in both strata, so that « R1 rejette (s) » [*R1 rejects (s)*] is stated for no stratum. A negative result is a result (doc 10 §7). The ASN axis is observable: 7 of the 10 hosts of the pool share AS13335 (Cloudflare), on the delivery side (J28 I.435691).

What it does not allow one to say. It neither establishes nor excludes a co-failure of the sources: the cause of the discordance is not identified (sealed statement). It establishes no pair dependence (ADR-0028 D3: descriptive). It does not establish the independence of the sources (doc 09). The pre-registered descriptives (§5), including the decomposition of K by `panne_transport` readings, are hors décision [*outside the decision*] and identify no cause. The hors décision outputs that print « R1 discrimine » VRAI [*true*] (main J14; « page incluse » [*range included*] variant of the third-party recomputation) do not change the verdict (§7). Nothing is said about the accuracy of a price, nor outside the measured axes (A(axis-coverage), doc 08).

Decisions of the investor (stated; this report settles none of them). Pre-registered options:

- ADR-0028 D6 (vi), quoted (inner quotation marks rendered as ' '): "Clause dated after J28: decision of the investor (§4.10 b), on the rule written in the package (SHOGEN-CRITERE-R1-1). If R1 discrimine: 'continuous benchmark' work package through a new G0 (reimplementation in the core, or G0-G7 promotion of the collection). Otherwise, the collection dies as planned." Dated erratum of the same paragraph: "'R1 discrimine' := VRAI of rule SHOGEN-CRITERE-R1-1; 'the collection dies' remains a decision of the investor".
- ADR-0028 D9, quoted: "After J28: bifurcation = decision of the investor (§4.10 b)", with, for the branch observed here: "If (i) is false: priority to G4 and revision of 04 (doc 10 I.35-36; 05-roadmap I.126-127)."
- Package §10.2 pt 11, quoted: "FAUX → priority to G4 and revision of 04, with the EMD_s "; "'The collection dies' remains a decision of the investor (D6 vi), never a mechanical consequence."
- Publication of J14 and of J28, and of this report (ADR-0028 §4 pt 4; D4: J14 is not published without a decision of the investor).

Decisions recorded since, after the execution and before this report (JOURNAL I.322, 2026-10-04 02:42:19 UTC; answers of the investor quoted): S2 collection, “Stop it (Recommended)”; S2-bis, “Yes, prepare S2-bis (Recommended)”; publication, “Yes, in principle (Recommended)”, the final go-ahead remaining with the investor after reading the report; one answer on commercial strategy is not reproduced. The JOURNAL notes there that the D9 bifurcation follows the branch “(i) false” (G4, revision of doc 04), plus an ADR-S2-BIS work package.

Reading conventions and lexicon

The renderings are cited by abbreviation and line number (“J28 I.435516” = line 435516 of the J28 rendering). All are in docs/adr-0028/execution/rendu-2026-10-04/, under the prefix shogen-27b0e30-rendu-20261004T011129Z-943.:

abbreviation	file	content	sha256
SUITE	0-suite.out	s2-harness suite (run suite)	4f223a22950778c799b39b1856a400c965ac18b4b57625bfef3136c04daeb3a
J14p	1-j14-principal.out	main J14, hors décision	246ebeeaa132392de15f34ccefff808e4e44b157498c3fdc1009282cc7e1e1498
J14s	2-j14-second.out	second J14, hors décision	d89db3a94da67f48c00f3bbc7ce7710a092fc3ad14f935e874e8c905f0172b24
J28	3-j28.out	confirmatory segment: blocks 1 to 6 and [SENSIBILITÉ] [SENSITIVITY]	26877549658b7271f766b433e941c79a35169398238755fa4bbf77a4911d65c0
RT	4-recalcul-tiers.out	third-party recomputation (recompute_*), JSON	a76ad44832d28751e68c56d6844be0c0ebd83f1b79b89bd7a16611779d891ac8
RAW	5-raw.out	verdict of the raw journal	e996c6343bbcd572221b6a3eb7679286167b46b3452431136f3d622254ae97b8
ENR	json	oracle record of role « rendu » [rendering]	355cf9bb43609c1c4b0f3ffebcd32d9d987e5e12e63277d9e4acbd467eb0fb5c

Other documents: PAQ = sealed package; ADR = docs/adr-0028/ADR-0028-decisions-sortie-S2.md; annexes A, B, D of the same folder; doc 04, doc 08, doc 09, doc 10 = docs/04-certificat-diversite.md, docs/08-assumptions.md, docs/09-vocabulaire.md, docs/10-mesures-pilotes-design.md.

- **Complete values:** Decimal strings (precision 50) copied as printed, between backquotes.
- **Reading roundings:** in the running text only, flagged by “≈”, to 3 significant figures; the complete value is in a table, with its line.
- **[calc]:** computation by the drafter (scripts d3_lift.py, controles_r1.py, comparer_recalcul.py, calc_divers.py, Python standard library, exact arithmetic or decimal; outputs cited in §8.9 and §6).
- **No p-value** is given or read (package §10.2 pt 4). The exact binomial tails that the J14 renderings print hors décision under the \$5.4 guard are not copied here; they remain readable at the lines cited (§7.1, §7.2).
- **Vocabulary:** register of doc 09. The independence of sources is never “established”: an independence model is rejected or not rejected, on n windows, on named axes, as observed by this instrument.

Lexicon, for an outside reader (definitions of doc 10 and of the package, summarized):

- **Feed, source, host:** a feed is a reading of the BTC/USD (or BTC/USDT) price served by a source (exchange, aggregator or oracle); two feeds can share a host (www.okx.com). The configured pool counts 12 feeds.
- **Window:** a minute of the UTC grid (w = 60 s); in each window, each feed counts as one reading.
- **Anomaly** (doc 10 §5.2): a feed is anomalous in a window if it is in outage (reading absent, status not ok or price absent), stale (carried timestamp older than σ_{classe}) or out of envelope (relative difference from the median of the other feeds beyond τ_{classe} , with at least four respondents).
- **n, K:** number of analysed windows of a stratum; number of those windows in which at least two feeds are anomalous.
- **P_{more}:** probability of at least two anomalies in a window if the feeds were unrelated to one another, estimated from the anomaly rate of each feed (Knight and Leveson model, doc 10 §5.1).
- **z_s, z_{bloc,s}:** difference between K and n-P_{more}, scaled by the binomial standard error with independent windows (z_s) or by a block standard error that allows a serial dependence of the windows of range below $l = 240$ windows, i.e. 4 h (z_{bloc,s}).
- **Rule SHOGEN-CRITERE-R1-1:** a tested stratum (guard n-P_{more}(1 - P_{more}) ≥ 10 met) REJETTE [rejects] if z_s ≥ 2.33 and z_{bloc,s} ≥ 2.33; « R1 discrimine » has the value VRAI if at least one stratum rejects.
- **Strata:** stress = Saturday and Sunday UTC; calm = the other days; calendar fixed before the campaign.
- **EMD_s:** minimum detectable excess of K, in windows, at power 0.8 (design choice).
- **R2, k_{eff}, nominal k:** diversity observables; here the ASN axis, the network that announces the address of each host; k_{eff} = number of classes after merging the hosts that share a measured ASN; nominal k = number of hosts.
- **J28, J14:** J28 is the confirmatory segment (38,600 distinct windows at fixed n, then D5 exclusion); the two J14 segments are cuts of the start of the campaign (fourteen days for the main one, up to 2026-09-04 for the second), hors décision.
- **D1, D5:** removal of the feeds without any ok reading (here Pyth); exclusion of a range of degraded harness, from 24 to 26 September 2026.
- **Hors décision:** printed and labelled output, which cannot change the verdict (package §10.2 pt 9).

2. Instrument and campaign

2.1 Fact class, pool, feeds and hosts

- Class: « BTC/USD-stable, devise marquée par flux » [BTC/USD-stable, currency marked per feed]; « pool configuré = 12 flux » [configured pool = 12 feeds]: binance, coinbase, kraken, okx_ticker, okx_index, bitstamp, gemini, bitfinex, coingecko, defillama, pyth, chainlink (J28 I.8-9); currency composition of the analysis pool: 9 USD / 2 USDT (I.46).
- Window w = 60 s (I.10); Decimal decimals to 50 digits (I.19); threshold z = 2.33 (I.21); guard n-P_{more}·(1-P_{more}) ≥ 10 (I.16, I.20); n_{min} = 4 respondents for the out-of-envelope axis (I.17).
- Thresholds per class, committed before the campaign (J28 I.12-14):

class (σ/τ)	feeds	σ _{classe} (s)	τ _{classe}
place_horodatee	coinbase, okx_ticker, okx_index, bitstamp, gemini	180.0	0.0045
sans_horodatage	binance, kraken, bitfinex	None (staleness not evaluated)	0.0045
agregateur	coingecko, defillama	1140.0	0.026
oracle_chainlink	chainlink	10695.0	0.0165
oracle_pyth	pyth	33.0	0.0015

- Printed anomaly definition (J28 I.435448): per window and per feed of the analysis pool, precedence outage > staleness > out-of-envelope; K = windows with at least two anomalies. Printed residual: staleness « fail-open » for binance, kraken and bitfinex, which carry no timestamp (I.435464).
- Hosts: the 11 feeds of the analysis pool are served by 10 distinct hosts (« k nominal = 10 (hôtes distincts du pool) » [nominal k = 10 (distinct hosts of the pool)], J28 I.435779); okx_index and okx_ticker share www.okx.com (I.435688); chainlink is read through the public RPC ethereum-rpc.publicnode.com (I.435685).

2.2 Analysis pool (ADR-0028 D1)

stratum	removal (case)	printed counts	nominal k_s	source
calm	pyth, case (a)	« ok = 0 / 24585 lectures, n = 24585 » [ok = 0 / 24585 readings, n = 24585]	« 10 sources / 11 flux » [10 sources / 11 feeds]	J28 I.42, I.44
stress	pyth, case (a)	« ok = 0 / 11397 lectures, n = 11397 » [ok = 0 / 11397 readings, n = 11397]	« 10 sources / 11 flux »	J28 I.43, I.45

Pyth is removed from R1, from L&M and from R2 in both strata (case (a): no ok reading in any stratum). The ADR carries the count over the whole journal: "0 ok out of 40,804 readings, HTTP 401 from 2026-08-26T19:00Z" (ADR I.25). No removal under case (b) is printed.

2.3 J28 segment (ADR-0028 D4, D2 pt 6)

- Segment: [1787770800 ; 1790558880] = [2026-08-26T19:00:00+00:00 ; 2026-09-28T01:28:00+00:00], half-open, same bound on window_start, ts and harness_ts, fixed n = 38,600 (J28 I.31).
- The journal counts 38,600 distinct windows: first 1787770800 (2026-08-26T19:00:00Z), last window_start 1790558820 (2026-09-28T01:27:00Z) (I.28); T_fin = 1790558820 + 60 = 1790558880 [calc].
- The segment removes nothing: « fenêtres distinctes (window_close) calme 0, stress 0 ; asn_attribution 0 ; clock_check 0 » [distinct windows (window_close) calm 0, stress 0; asn_attribution 0; clock_check 0] (I.32).

2.4 D5 exclusion

- Bounds: window_start in [1790273880 ; 1790435280] (closed; 2026-09-24T18:18:00Z to 2026-09-26T15:08:00Z, J28 I.33); ts and harness_ts in [1790273880 ; 1790435340] (I.34).
- Removed by the range (I.35-36): 1,709 calm windows and 909 stress windows; 5,313 asn_attribution probe records (4,498 ok, 815 resolve_failed); 483 clock_check probe records.
- Reason (ADR I.60): resolve_failed 15.3% in the range against 1.85% outside the range, health of the DNS harness, not a source statistic; « causalité non établie » [causality not established] (recalled by the rendering, J28 I.435801).
- Check [calc]: 1,709 + 909 = 2,618; 38,600 – 2,618 = 35,982 = 24,585 + 11,397.

2.5 Strata and n per stratum

- Ex ante calendar: « kind=weekend_utc stress = samedi+dimanche UTC » [kind=weekend_utc stress = Saturday+Sunday UTC], operator decision committed before the launch, never inferred from the data (J28 I.27).
- n_calme = 24,585 completed windows (J28 I.435451); n_stress = 11,397 (I.435471).
- Coverage of the five weekends (J28 I.435814-435818): 2026-08-29, 1,140 windows out of 2,880 (19 h 00 min); 2026-09-05, 2,801; 2026-09-12, 2,611; 2026-09-19, 2,874; 2026-09-26, 1,971 in the excluded variant (32 h 51 min), 909 windows removed by the D5 range.

2.6 Skipped windows

- Grid with a 60 s step over the segment, without a window_close marker, outside the D5 range: 5,701 in calm, 2,094 in stress, all causes combined, under the assumption H_perte (A(loss-non-informative), doc 08) (J28 I.37).
- Breakdown (I.38-41): « harnais vivant » [harness alive] (skipped between two markers of the same start) 17 in calm, 150 in stress; cause not attributed by the journal (stop or transition between starts) 5,684 in calm, 1,944 in stress.
- Check [calc]: the grid of the segment counts (1790558880 – 1787770800)/60 = 46,468 windows; 46,468 – 38,600 (markers) – 73 (windows of the range without a marker, J28 I.435806) = 7,795 = 5,701 + 2,094.
- The harness started 4,093 times over the campaign, with run_params concordant on the load-bearing fields (J28 I.26).

2.7 Collection host and limitations of the observer

- The S2 collection is a bare HTTPS reading by a harness (doc 10 §1, non-goal 1); the only interaction with a chain is an eth_call reading through public RPC (doc 10 §1, non-goal 4).
- The renderings carry the trace of a Windows collection host: path with a drive letter (J28 I.7); socket errors WinError 10013 in the ASN probe records at the end of the main J14 (J14p I.209885). The JOURNAL places this collection on the personal workstation of the investor, who asks for a collection on a virtual private server for S2-bis (JOURNAL I.324).
- A single observation point and a single resolver per ASN probe record, cloudflare-dns.com (J28 I.435676-435688); the rendering itself publishes residuals 3 (single-vantage; multi-resolver probe « DUE », J28 I.435772) and 5 (the measurement path belongs to the measured class, « le résidu est à son MAXIMUM, publié » [the residual is at its MAXIMUM, published], I.435774).
- The assignment of windows to strata and segments rests on the host's clock, A(harness-clock) (doc 08; package §12 pt 17). Block 1 of J28 prints 3,610 start-up controle_horloge probe records (I.48-3657; count [calc] by line search); their range is not computed by the rendering (package §12 pt 17) nor by this report.
- The label of the annex D.5 descriptive recalls it: « le z confirmatoire inclut les modes communs de l'observateur (hôte, DNS, réseau) » [the confirmatory z includes the common modes of the observer (host, DNS, network)] (J28 I.435531).

3. Confirmatory verdict (J28, rule SHOGEN-CRITERE-R1-1)

The rule is evaluated mechanically by the single-execution script on the J28 segment, D1 analysis pool per stratum, D5 exclusion, in each confirmatory stratum (package §10.2 pt 1). The comparisons bear on the unrounded Decimal values, against the constant Decimal("2.33"), with the inequality « ≥ » (pt 4; printed reminder, J28 I.435515).

3.1 Inputs and value per stratum (complete values)

input (pt 1)	calm	stress	J28
n_s (completed windows)	24585	11397	I.435451; I.435471
K_s (windows with ≥ 2 anomalies)	154	133	I.435451; I.435471
$\hat{P}_0$	0.94236061369788154146024296580093086828513938470018	0.94068072247707197570660265774160717593176388308751	I.435465; I.435485
$\hat{P}_1$	0.056276435879320695626255997808433611118939114709424	0.057852878561437609992981152213024246102249979400096	I.435466; I.435486
$\hat{P}_{\text{more},s}$	0.0013629504227977629135010363906355205959215005903986	0.0014663989614904143004161900453685779659861375123928	I.435467; I.435487
guard n_s· $\hat{P}_{\text{more},s}(1-\hat{P}_{\text{more},s})$ (threshold 10)	33.462466216157713120610261901717310599417168320476	16.688041699661428674928415017858500902580981925520	I.435468; I.435488
z_s	20.829495417808173515660354538360738280133057485734	28.466243694685921454102240881955034913016073716310	I.435469; I.435489
ℓ	240	240	I.435494; I.435498
$\hat{Y}_{0,s}$	153.03534675615212527964205816554809843400447427293	131.44792489251557427393173642186540317627445819075	I.435494; I.435498
$\hat{\sigma}^2_{\text{bloc},s}$	3837.3644239148386709307388556071047850697416032311	4444.2268779663177986090438425985366511755710614584	I.435494; I.435498
FIV_s	114.67667682132543551816351669459307932477895917439	266.31206692493364443057277635430175333060222847741	I.435495; I.435499
R_centrage,s	4.573343324247139833425606396397303313682246279750	7.8767735159232268649296718039283324323387017357003	I.435495; I.435499
FIV_série,s	25.075020282924108215938185508697612502356118056163	33.809791075822184156289564915103491737231918165960	I.435495; I.435499
z_bloc,s	1.9450967130759827973431520036612640074344098711768	1.7443544612798118150330419847751992092506597257981	I.435496; I.435500
maximal run of I_t	6	2	I.435497; I.435501
EMD_s (windows)	196.46940578076511214605312297686775770638678422511	211.43482468284790300599634573463971654187425170170	I.435518; I.435521
fraction EMD_s/n_s	0.0079914340362320566258309181605396688105099363117799	0.018551796497573738966920798958904950122126371124129	I.435518; I.435521
rule value (pt 5)	NE REJETTE PAS (discordance)	NE REJETTE PAS (discordance)	I.435516; I.435519

Reading (roundings flagged): in calm, $n = 24,585$, $K = 154$, $\hat{P}_{\text{more}} \approx 0.00136$, guard ≈ 33.5 (met), $z_s \approx 20.8 \geq 2.33$ and $z_{\text{bloc}} \approx 1.95 < 2.33$; in stress, $n = 11,397$, $K = 133$, $\hat{P}_{\text{more}} \approx 0.00147$, guard ≈ 16.7 (met), $z_s \approx 28.5 \geq 2.33$ and $z_{\text{bloc}} \approx 1.74 < 2.33$. Both strata are tested: $\$5.4$ guard met, $\hat{\sigma}^2_{\text{bloc},s} > 0$ and $n_s \geq 30 \cdot \ell = 7,200$, hence $z_{\text{bloc},s}$ published (pts 2, 3 and 5).

[calc] By construction (same numerator $K_s - n_s \cdot \hat{P}_{\text{more},s}$, package pt 3), $z_{\text{bloc},s} = z_s / \sqrt{FIV_s}$: the difference between the two statistics is carried entirely by $FIV_s \approx 115$ in calm and ≈ 266 in stress, product of $R_{\text{centrage},s}$ (≈ 4.57 ; ≈ 7.88) and of $FIV_{\text{série},s}$ (≈ 25.1 ; ≈ 33.8). The report draws no cause from it (pt 8).

3.2 Printed statements (pt 8), word for word

Lines J28 I.435515 to I.435522, copied by extraction (sed -n '435515,435522p'):

— règle SHOGEN-CRITERE-R1-1 (ADR-0028 §1 bis.1 pts 1-11 ; forme scellée) : valeur par strate ; comparaisons sur les Decimal publiées, non arrondies, au seuil 2.33, « $\geq$ » ; aucune p-valeur
« calme » : $z_s = 20.829495417808173515660354538360738280133057485734 \geq 2,33$; $z_{\text{bloc}} = 1.9450967130759827973431520036612640074344098711768 < 2,33 \rightarrow$ NE REJETTE PAS (discordance)
« calme » : « le modèle binomial de doc 10 §5.1, à fenêtres indépendantes, est rejeté ; la cause n'est pas identifiée entre co-défaillance de s sources et dépendance sérielle des fenêtres »
« calme » : $EMD_s = (2,33 + 0,8416) \cdot \max(\sqrt{(n_s \cdot \hat{P}_{\text{more},s} \cdot (1 - \hat{P}_{\text{more},s}))}, \sigma_{\text{bloc},s}) = 196.46940578076511214605312297686775770638678422511$ fenêtres ; fraction de $n_s = 0.0079914340362320566258309181605396688105099363117799$ (puissance 0,8 : choix de conception ; aucun seuil sur l'EMD)
« stress » : $z_s = 28.466243694685921454102240881955034913016073716310 \geq 2,33$; $z_{\text{bloc}} = 1.7443544612798118150330419847751992092506597257981 < 2,33 \rightarrow$ NE REJETTE PAS (discordance)
« stress » : « le modèle binomial de doc 10 §5.1, à fenêtres indépendantes, est rejeté ; la cause n'est pas identifiée entre co-défaillance de s sources et dépendance sérielle des fenêtres »
« stress » : $EMD_s = (2,33 + 0,8416) \cdot \max(\sqrt{(n_s \cdot \hat{P}_{\text{more},s} \cdot (1 - \hat{P}_{\text{more},s}))}, \sigma_{\text{bloc},s}) = 211.43482468284790300599634573463971654187425170170$ fenêtres ; fraction de $n_s = 0.018551796497573738966920798958904950122126371124129$ (puissance 0,8 : choix de conception ; aucun seuil sur l'EMD)
« R1 discrimine » (§1 bis.1 pt 6 ; déclencheur de D6 (vi) et D9) = FAUX : aucune strate ne rejette ; strate(s) testée(s) : calme, stress

The discordance case (guard met, $z_s \geq 2.33$, published $z_{\text{bloc},s} < 2.33$) has the value NE REJETTE PAS; the stratum prints the discordance statement and EMD_s, and not the statement « n'est pas rejeté » [*is not rejected*]; this case is never a VRAI (package §10.2 pts 5 and 8). This “single test” reading of the discordance was ratified by the orchestrator on 2026-09-30, subject to veto, before the sealing (package §10.1). $EMD_s = (2.33 + 0.8416) \cdot \max(\sqrt{(n_s \cdot \hat{P}_{\text{more},s}(1 - \hat{P}_{\text{more},s}))}, \hat{\sigma}_{\text{bloc},s})$: the power 0.8 is a design choice, without any threshold on the EMD (pt 8). Here the maximum is $\hat{\sigma}_{\text{bloc},s}$ in both strata [calc: $\sqrt{\hat{\sigma}^2_{\text{bloc},s}} \approx 61.9$ and ≈ 66.7 against $\sqrt{\text{guard}} \approx 5.78$ and ≈ 4.09].

3.3 « R1 discrimine »

« R1 discrimine » has the value FAUX: « aucune strate ne rejette ; strate(s) testée(s) : calme, stress » [*no stratum rejects; stratum(s) tested: calm, stress*] (J28 I.435522). By pt 6, FAUX means that no stratum REJETTE, that at least one NE REJETTE PAS and that none is in a non-qualifiable rejection. It is the trigger of D6 (vi) and of D9 (§1, decisions of the investor).

3.4 Bonferroni family and premise of the bound (pt 7)

Printed line (J28 I.435503):

famille de Bonferroni pré-enregistrée (ADR-0028 D2 pt 4) : m = 2 (strates testées : calme, stress ; m ≤ 2 ; §1 bis.1 pt 6), tests unilatéraux a u seuil 2,33 ; borne P(au moins un rejet à tort) ≤ 2 × 0,01 = 0,02 sous le modèle nul joint (§1 bis.1 pt 7) : modèle d'indépendance du pool de do c 10 §5.1 et dépendance sérielle des fenêtres de portée < ℓ = 240 (A(window-dependence), registre 08) ; niveau asymptotique, non démontré ≤ 0,01 en échantillon fini (SHOGEN-SIM-NIVEAU-1)

Premise, quoted from package §10.2 pt 7: “The bound holds under the joint null model: the independence model of the pool of doc 10 §5.1, with a serial dependence of the windows of range < ℓ (A(window-dependence), doc 08). The level is asymptotic [...]. It is conservative under iid windows (plug-in of P_more). It is not demonstrated ≤ 0.01 in finite samples”. The measurements of SIM-NIVEAU and of step S, made before the sealing on synthetic data only (package §10.3), say nothing about the real sources: under the simulated null model, the false-rejection rate of the rule ranges from 0.001590 (SE 0.000126) to 0.001810 (SE 0.000134) with independent windows, and stays at most 0.00357 (SE 0.00034) near the guard; that of z_s alone reaches 0.337300 (SE 0.001495) under serial dependence (mean runs of 60 windows) (package §10.3, tables I.139-148 and I.153-162). By pre-declared consequence, these outputs modify neither ℓ, nor the threshold, nor the rule (pt 7).

3.5 Anomalies per feed (block 3)

⧘_i = anomalies of the feed / n_s; the complete values of ⧘_i are at the lines cited. « nonÉv » [not evaluable] counts the cells not evaluable on the out-of-envelope axis, which are not anomalies (J28 I.435448).

feed	class	calm: anomalies (panne / stale / horsE / nonÉv) [outage / stale / out-of-envelope / not evaluable]	stress: anomalies (panne / stale / horsE / nonÉv)	J28
binance	sans_horodatage	372 (372 / 0 / 0 / 0)	46 (46 / 0 / 0 / 0)	I.435453; I.435473
coinbase	place_horodatee	56 (56 / 0 / 0 / 0)	11 (11 / 0 / 0 / 0)	I.435454; I.435474
kraken	sans_horodatage	34 (34 / 0 / 0 / 0)	46 (46 / 0 / 0 / 0)	I.435455; I.435475
okx_ticker	place_horodatee	64 (64 / 0 / 0 / 0)	6 (6 / 0 / 0 / 1)	I.435456; I.435476
okx_index	place_horodatee	64 (64 / 0 / 0 / 0)	7 (7 / 0 / 0 / 1)	I.435457; I.435477
bitstamp	place_horodatee	475 (475 / 0 / 0 / 0)	39 (39 / 0 / 0 / 0)	I.435458; I.435478
gemini	place_horodatee	68 (65 / 3 / 0 / 0)	21 (19 / 2 / 0 / 0)	I.435459; I.435479
bitfinex	sans_horodatage	32 (30 / 0 / 2 / 0)	51 (51 / 0 / 0 / 0)	I.435460; I.435480
coingecko	agregateur	56 (56 / 0 / 0 / 0)	127 (127 / 0 / 0 / 1)	I.435461; I.435481
defillama	agregateur	159 (158 / 1 / 0 / 0)	226 (226 / 0 / 0 / 0)	I.435462; I.435482
chainlink	oracle_chainlink	71 (69 / 0 / 2 / 0)	113 (113 / 0 / 0 / 0)	I.435463; I.435483

4. Flags and R2 (blocks 5 and 6 of J28)

4.1 Certificate head (block 6), copied

Lines J28 I.435779 to I.435797, copied by extraction (sed -n '435779,435797p'):

```
k nominal = 10 (hôtes distincts du pool)
k_eff    = 4 - k_eff = nombre de classes de la partition R2 measured (§5.6)
date de la partition, axe ASN (ADR-0026 déc. 1 ; HS2-07) = relevé asn_attribution retenu (dernier par hôte, après filtre de lecture) : min 1790
558275.4337332 = 2026-09-28T01:17:55.433733+00:00 ; max 1790558282.5194278 = 2026-09-28T01:18:02.519428+00:00 ; 10 / 10 hôtes du pool – descripti
f seulement (ADR-0028 annexe D.5)
relevés asn_attribution retenus sans ts (hôtes du pool, entrée malformée) = 0 – comptés à part, hors date de la partition (SHOGEN-BLOC6-TS-1)
R3 (déclaration : entité légale, juridiction, méthodologie annoncée) ne modifie JAMAIS k_eff (§5.6 / 04 §3) – seuls les recouvrements R2 mesur
ed partitionnent.
PARTITION NOMMÉE (ADR-0007 : nomme l'amont, jamais un compte anonyme) :
cluster = {api.pub.bitfinex.com, api.coingecko.com, api.exchange.coinbase.com, api.kraken.com, coins.llama.fi, ethereum-rpc.publicnode.com, w
ww.okx.com} via AS13335 CLOUDFLARENET - Cloudflare, Inc. [fusion ASN partagé AS13335 CLOUDFLARENET - Cloudflare, Inc. – côté livraison (§4.1, rés
idu l)] ; flux = ['bitfinex', 'chainlink', 'coinbase', 'coingecko', 'defillama', 'kraken', 'okx_index', 'okx_ticker']
  amont déclaré (basis:doc, ne fusionne pas) : binance → coingecko (coingecko.com/en/exchanges/binance)
  amont déclaré (basis:doc, ne fusionne pas) : coinbase → pyth (pyth.network/publishers)
  amont déclaré (basis:doc, ne fusionne pas) : coingecko → defillama (docs.llama.fi/llms-full.txt)
  caveat : chemin de LECTURE ≠ amont (§3.2) : l'ASN mesuré est celui du fournisseur RPC (ethereum-rpc.publicnode.com), une infrastructure d
e lecture, pas l'amont du feed
cluster = {api.binance.com} via AS16509 AMAZON-02 - Amazon.com, Inc. ; flux = ['binance']
  amont déclaré (basis:doc, ne fusionne pas) : binance → coingecko (coingecko.com/en/exchanges/binance)
cluster = {api.gemini.com} via AS14618 AMAZON-AES - Amazon.com, Inc. ; flux = ['gemini']
cluster = {www.bitstamp.net} via AS19551 INCAPSULA - Incapsula Inc ; flux = ['bitstamp']
DRAPEAU 1 « historique insuffisant » (§5.4) par strate : {'calme': False, 'stress': False}
DRAPEAU 2 « co-défaillance observée non expliquée par les axes R2 » (§5.6) : état = ETEINT
« R1 discrimine » FAUX : le modèle d'indépendance n'est rejeté dans aucune strate testée (bloc 3)
entrées (ADR-0028 §1 bis.1 pt 10) : « R1 discrimine » = FAUX (bloc 3) ; k_eff = 4 ; k nominal du segment (hôtes) = 10 ; k nominal_s (flux d
u pool de la strate) : « calme » = 11, « stress » = 11 – comparaison hétérogène déclarée ; strate poolée hors des entrées
```

- **Named and dated partition:** four classes on the ASN axis, including a cluster of seven hosts and eight feeds merged on AS13335 (Cloudflare), « côté livraison » [*delivery side*] (residual 1 of doc 10 §4.1), and three singletons: api.binance.com (AS16509), api.gemini.com (AS14618), www.bitstamp.net (AS19551). Date: retained asn_attribution probe records, last per host, between 2026-09-28T01:17:55Z and 01:18:02Z, 10 hosts out of 10 (I.435781). The basis:doc edges (binance → coingecko, coinbase → pyth, coingecko → defillama) are declared upstreams: they do not merge (ADR-0008, recalled at I.435786-435788).
- **k_eff = 4, nominal k = 10** (distinct hosts of the pool, I.435779-435780); nominal k_s = 11 feeds in each stratum, « comparaison hétérogène déclarée » [*heterogeneous comparison declared*] (I.435797). k_eff is not an independence count: it is the number of classes of the observed R2 partition (doc 10 §5.6); z is never composed into k_eff.
- **Flag 1** « historique insuffisant » [*insufficient history*]: false in both strata (I.435794).
- **Flag 2** « co-défaillance observée non expliquée par les axes R2 » [*co-failure observed, not explained by the R2 axes*]: **ÉTEINT** [off] (I.435795). By pt 10, the flag is off as soon as « R1 discrimine » is FAUX; its inputs are printed (I.435797), pooled stratum excluded. The flag is a signal on A(axis-coverage), never the rule (pt 10).

4.2 R2 in summary (block 5)

- **(a) ASN axis** (I.435673-435691): 10 hosts, one retained probe record each, the last one (2026-09-28T01:17:55Z to 01:18:02Z), resolver cloudflare-dns.com, cross attribution RIPEstat and Team Cymru « concordant » for the 10. One measured overlap: AS13335 shared by api-pub.bitfinex.com, api.coingecko.com, api.exchange.coinbase.com, api.kraken.com, coins.llama.fi, ethereum-rpc.publicnode.com and www.okx.com (I.435691). Printed CNAME chains: binance to CloudFront, gemini to an AWS load balancer, bitstamp to Incapsula, okx to Cloudflare's CDN (I.435678, I.435682, I.435687, I.435689).
- **(b) Content axis** (I.435693-435752): 35,982 windows; N_min = 300 common windows per pair (design choice); sole merging criterion v0: exact identity (T = 1) on at least N_min common windows (I.435695). No pair is merged on this axis: the list of exact copies of the third-party recomputation is empty (RT I.10991). The per-pair statistics (p_raw, p_resid, T, T_Δ, co-aberrance, lag) are descriptive; the USDT/USD peg gap is carried by p_resid (J28 I.47). The serial dependence on this axis is out of scope (SHOGEN-CONTENU-DEP-1, §11).
- **(c) Method axis** (I.435754-435764): five basis:doc entries (three upstream edges: binance → coingecko, coinbase → pyth, coingecko → defillama; two estimator declarations: coingecko, pyth), which trigger (b) and do not partition.
- **(d) Correlations between clusters:** the rendering prints « aucun cluster à ≥ 2 flux (1) → tout reste au ∅ par paire de flux (bloc 4) » [*no cluster with ≥ 2 feeds (1) → everything stays at the ∅ per pair of feeds (block 4)*] (I.435767). The JSON of the third-party recomputation carries, for J28, "cluster_pairs": {} and "n_multi_clusters": 1 (RT I.10985-10987). [report finding] The label reads poorly next to block 6, which names a cluster of eight feeds: the partition counts a single cluster of at least two members, hence no pair of such clusters, and the main D3 estimator between clusters (correlation of the Θ series, analogue of eq. 35 of L&M) has no object on J28.
- **(e) The seven residuals of the ASN axis** are printed with the table (I.435769-435776), notably: fronting (k_eff counted on the delivery side), single-vantage, instantaneity (dated attribution, re-measured at each quorum), and the measurement path that belongs to the measured class.

4.3 ASN divergences for hosts outside the pool (SHOGEN-ASN-DIVERGENCE-HORS-POOL-1, annex B.44)

No ASN divergence appears in the renderings. A search for the string DIVERGENCE gives 0 lines in J14p, J14s, J28, RT and RAW, and a single one in SUITE, which is a test name (SUITE I.379); the JSON of the third-party recomputation carries "asn_divergences": [] in its four entries (RT I.4522, I.8900, I.13335, I.18099). There is therefore no line to label or to remove: the case that the item anticipated (probe records of hosts removed by D1 case (a), that is, of Pyth) does not arise in these renderings.

4.4 Caveats of the reading path (RPC)

- Chainlink is read by eth_call on the public RPC ethereum-rpc.publicnode.com (doc 10 §1, non-goal 4; J28 I.435685). The rendering says so: « chemin de LECTURE ≠ amount (§3.2) : l'ASN mesuré est celui du fournisseur RPC (ethereum-rpc.publicnode.com), une infrastructure de lecture, pas l'amount du feed » [*READING path ≠ upstream (§3.2): the measured ASN is that of the RPC provider (ethereum-rpc.publicnode.com), a reading infrastructure, not the upstream of the feed*] (I.435789). The classification of chainlink in the AS13335 cluster therefore bears on the reading infrastructure.
- Same rule for any host served behind a CDN: the observed IP is the delivery layer, not the origin (residual 1, I.435770).
- Pyth, read through an HTTP service that answered 401 from the first day (ADR I.25), is outside the analysis pool (§2.2): no measurement of this report bears on it.

5. Pre-registered descriptives (annex D.5, package §11), hors décision

These treatments are descriptive, hors décision and without parameters; none enters the rule or the closed list of sensitivities (package §11; printed label, J28 I.435524).

5.1 Observed τ against committed τ (SHOGEN-TAU-REDERIV-1)

The τ committed before the campaign serves all the confirmatory tests; the observed τ is defined in advance: P99 and maximum of |p – médiane_LOO|/médiane_LOO of the (window, feed) cells that reached the out-of-envelope axis, per class, over the segment, without re-estimation (J28 I.435525; package §11).

class	τ classe (committed)	N cells	observed P99	observed maximum
agregateur	0.026	71395	0.0029870446819581321147177994146356252452815395638463	0.017649908208186827670019929236496737002324764533841
oracle_chainlink	0.0165	35800	0.0045133817598667881395689083527158792588820593853894	0.017842859661860400175061719273882056236385914669301
oracle_pyth	0.0015	0	not defined	not defined
place_horodatee	0.0045	179097	0.00084986241659292805372128990214331880618269790009557	0.003100597179657658675816929027355805819826557001381
sans_horodatage	0.0045	107367	0.0016680227037502354455795456531406013037374194387754	0.008429372292219192397562143693388819212812006552111

[calc] The observed P99 is below τ_classe in the four defined classes (P99/τ ratios ≈ 0.115; 0.274; 0.189; 0.371). The observed maximum exceeds τ_classe for oracle_chainlink (≈ 1.08 τ) and sans_horodatage (≈ 1.87 τ), which is consistent with the out-of-envelope cells counted in block 3 (chainlink and bitfinex, 2 each in calm, §3.5). A divergence between observed τ and projected τ is a finding, not a silent re-tune (ADR-0022 pt 3, cited by package §12 pt 13); the re-derivation is PX-Shogen-13, after the rendering, never applied to the confirmatory z. Reservation already written (annex B.45, note to PX-Shogen-13): the observed τ (out-of-envelope axis reached) and the calibration P99 (all evaluable cells) do not bear on the same population.

5.2 Decomposition of K by panne_transport readings (SHOGEN-HOST-DEGRADED-1)

Lines J28 I.435531 to I.435533, copied by extraction:

```
décomposition de K (SHOGEN-HOST-DEGRADED-1) par nombre de lectures présentes au statut panne_transport (model.Status) des flux du pool D1 de la strate ; c_s = fenêtres où chaque flux du pool porte une lecture panne_transport ; le z confirmatoire inclut les modes communs de l'observateur (hôte, DNS, réseau)
« calme » : K = 154 = K[≥ 2 panne_transport] 153 + K[1] 1 + K[0] 0 ; K[tous les écarts hors_enveloppe] = 0 ; c_s = 9
« stress » : K = 133 = K[≥ 2 panne_transport] 133 + K[1] 0 + K[0] 0 ; K[tous les écarts hors_enveloppe] = 0 ; c_s = 0
```

In calm, 153 of the 154 windows counted in K carry at least two readings with status panne_transport among the feeds of the stratum's pool, one carries one, none carries zero; in stress, the 133 carry at least two. No window of K has all its anomalies on the out-of-envelope axis. c_s, the number of windows in which each feed of the pool carries a panne_transport reading, is 9 in calm and 0 in stress. This descriptive identifies no cause: it describes the composition of K, which the rule does not read. The sensitivity that removes the windows with a degraded-host diagnostic is a post-execution item, not done at the date of the report (SHOGEN-HOST-DEGRADED-2, §11).

5.3 Censoring: skipped windows and non-outer bounds (SHOGEN-CENSURE-INFO-1)

The skipped windows are counted in §2.6 (J28 I.37-41). Under the assumption H_perte (A(loss-non-informative): tooling losses are non-informative), the rendering prints bounds at fixed $\hat{P}_{more}$, with s = skipped windows of the stratum: $z_{bas} = (K - (n+s) \cdot \hat{P}) / \sqrt{((n+s) \cdot \hat{P}(1-\hat{P}))}$ and $z_{haut} = (K + s - (n+s) \cdot \hat{P}) / \sqrt{((n+s) \cdot \hat{P}(1-\hat{P}))}$, then the same with $\hat{\sigma}_{bloc,s}$ in the denominator (J28 I.435535). Printed header line: « bornes à $\hat{P}_{more}$ fixé, non extérieures ; verdict non identifié sous censure arbitraire des fenêtres sautées » [bounds at fixed $\hat{P}_{more}$, non-outer; verdict not identified under arbitrary censoring of the skipped windows] (I.435534).

stratum	s (skipped windows)	z_bas	z_haut	z_bas
calm	5701	17.556689997555283653681146237762934756839540547496	905.50199539544630203447432173216643843074844907832	1.819662913686122047171
stress	2094	25.473078564141066684303998709584385344769540455886	496.61005684969625124835679931557781942595489865145	1.698293742492909177994

These bounds are labelled « non extérieures » [non-outer] and carry no identification reading (annex D.5; CV2-24): imputing an anomaly to a censored window also raises $\hat{P}_{more}$, so that the lower bound is not outer. The loss counts measure the extent of the censoring, not its non-informative character (package §12 pt 14). The outer bounds are item SHOGEN-CENSURE-INFO-2, after the execution (§11).

5.4 Diagnostic of runs of I_t and FIV

- Runs of $I_t = 1\{m_t \geq 2\}$ (hors décision; J28 I.435497, I.435501): in calm, 119 runs, mean length 1.2941176470588235294117647058823529411764705882353, maximal run 6; in stress, 130 runs, mean length 1.0230769230769230769230769230769230769230769230769230769230769230769, maximal run 2. [calc] The mean length is K_s divided by the number of runs (154/119 and 133/130).
- The maximal run stays below $l = 240$ in both strata: the flag « run maximal $\geq l$: $\hat{\sigma}_{bloc,s}$ biaisé vers le bas ; SHOGEN-DEP-FENETRES-2 prioritaire avant G10 » [maximal run $\geq l$: $\hat{\sigma}_{bloc,s}$ biased downward; SHOGEN-DEP-FENETRES-2 a priority before G10] (pt 9) is not printed; in J28, the phrase appears only in the head label (I.1).
- FIV_s and its two factors are in §3.1 (I.435495, I.435499); the theoretical coefficient of variation of $\hat{\sigma}_{bloc,s}$, $\sqrt{4/(3n)}$, is 0.11408797792643129814123654880491384644099142081780 in calm and 0.16756361261114975367661284639462247100481161298594 in stress (I.435494, I.435498), marked « [inféré : dérivation de l'AVIS-advisor-defi Q1 (iv)] » [[inferred: derivation from the AVIS-advisor-defi Q1 (iv)]] by the rendering (I.435493).

6. Lift and ϕ identity of D3 (SHOGEN-D3-LIFT-1, annex B.42)

D3 (package §5, quoted in substance): between singletons of the R2 partition, ϕ of the anomaly indicators remains the main estimator; for the pairs of singletons (2x2 tables), the lift (left-hand side of eq. 28 of L&M) is a declared secondary; “No significance per pair: this is descriptive”; the identity $\phi = (lift-1) \cdot \sqrt{p_A p_B / ((1-p_A)(1-p_B))}$ must be published, “which makes the mention ‘noisy proxy’ exact”, and the four counts per pair. The renderings do not compute the lift (finding B-2 of review R-B, annex B.42): it is recomputed here by the drafter, in exact arithmetic, from the four counts printed in block 4 of J28 alone.

- Pairs concerned:** the J28 partition counts three singletons, {binance}, {gemini} and {bitstamp} (J28 I.435790, I.435792, I.435793), hence three pairs of singletons per stratum.
- Definitions:** $n = n_{11} + n_{10} + n_{01} + n_{00}$; $p_A = (n_{11} + n_{10})/n$; $p_B = (n_{11} + n_{01})/n$; $lift = p_{AB}/(p_A p_B) = n_{11} \cdot n / ((n_{11} + n_{10})(n_{11} + n_{01}))$; $\phi = (n_{11} \cdot n_{00} - n_{10} \cdot n_{01}) / \sqrt{((n_{11} + n_{10})(n_{01} + n_{00})(n_{11} + n_{01})(n_{10} + n_{00}))}$.
- Method:** script d3_lift.py (standard library, fractions and decimal to 80 digits), filed in the project repository with this report (sha256 8f91474cdedd5694e733334f4df225b110d3378ddb560f1bee4066eca80f9381; output d3_lift.sortie.txt, sha256 77ae69ec8666065d0e7ec3a13588f982d87de7bf1789ffe3f91813ea028dc73f). The identity is checked exactly: ϕ^2 and $(lift-1)^2 \cdot p_A p_B / ((1-p_A)(1-p_B))$ are two equal rationals, and ϕ has the sign of $lift-1$. Self-test on two tables computed by hand ([2 1 1 6]: $lift = 20/9$, $\phi = 11/21$; [0 5 5 90]: $lift = 0$, $\phi = -1/19$), which fails under a mutated identity.

[calc] Output of the script, put in a table (table_lift.py, string rewriting); the exact lift is an irreducible fraction, its decimals are rounded to 12 significant figures:

stratum	pair	[n11 n10 n01 n00] (J28)	p_A; p_B	lift (exact)	lift $\approx$	$\sqrt{p_A p_B / ((1-p_A)(1-p_B))} \approx$	(lift - 1) · $\sqrt{(...)} \approx$	printed ϕ (block 4)
calm	binance×bitstamp	[52 320 423 23790] (I.435547)	124/8195; 95/4917	21307/2945	7.23497453311	0.0173978414435	0.108475098331	0.10847509833108362918259591106802162
calm	bitstamp×gemini	[21 454 47 24063] (I.435551)	95/4917; 68/24585	103257/6460	15.9840557276	0.00739211972867	0.110763933959	0.11076393395917706370581636186843800
calm	binance×gemini	[20 352 48 24165] (I.435552)	124/8195; 68/24585	40975/2108	19.4378557875	0.00652781686160	0.120358945901	0.12035894590128557980983967111642447

stratum	pair	[n11 n10 n01 n00] (J28)	p_A; p_B	lift (exact)	lift ≈	$\sqrt{(p_A \cdot p_B / ((1 - p_A)(1 - p_B)))} \approx$	$(\text{lift} - 1) \cdot \sqrt{(\dots)} \approx$	printed ϕ (block 4)
stress	binance×bitstamp	[4 42 35 11316] (1.435622)	46/11397; 13/3799	7598/299	25.4113712375	0.00373029540546	0.0910616259682	0.09106162596816821457955675280253596
stress	binance×gemini	[3 43 18 11333] (1.435624)	46/11397; 7/3799	11397/322	35.3944099379	0.00273512204339	0.0940729087904	0.09407290879042906189572135484359554
stress	bitstamp×gemini	[3 36 18 11340] (1.435627)	13/3799; 7/3799	3799/91	41.7472527473	0.00251765505523	0.102587526866	0.10258752686577574340266708561888701

In rounded reading [calc]: lift ≈ 7.23; 16.0; 19.4 in calm and ≈ 25.4; 35.4; 41.7 in stress, for printed ϕ from ≈ 0.108 to ≈ 0.120 in calm and from ≈ 0.0911 to ≈ 0.103 in stress. The factor $\sqrt{(p_A \cdot p_B / ((1 - p_A)(1 - p_B)))}$ is small at low margins (from ≈ 0.00252 to ≈ 0.0174 here), and lift − 1 equals ϕ divided by this factor: this is the relation that the package publishes to make the mention “noisy proxy” exact. No decision reading is drawn from it, no per-pair significance is computed, and the conditional odds ratio, exploratory and without a method fixed at the sealing, is not computed (package §5).

Check of the 110 tables of block 4 [calc]: the identity is exact on 110 tables out of 110 (55 per stratum), and the recomputed ϕ differs from the printed ϕ by at most $2.38 \cdot 10^{-50}$ in absolute value (output d3_lift.sortie.txt, l.2-3). The lift is published only for the pairs of singletons, the only ones targeted by D3.

7. Outputs hors décision (package §10.2 pt 9)

Pt 9 lists them: main and second J14 (non-confirmatory), pooled stratum (exploratory), sensitivities « plage incluse » (biased upward by construction) and « seconde coupe » [second cut], L&M (block 4), exact tails, run diagnostic. **None changes the verdict:** the rule is evaluated on J28 alone, D5 range excluded (pt 1), and « R1 discrimine » there has the value FAUX (§3.3). Each output is reproduced with the label that the rendering prints at its head.

7.1 Main J14

Label (J14p l.1):

[ÉTIQUETTE] j14-principal : hors décision, non confirmatoire (ADR-0028 D4, §1 bis.1 pt 9) ; plage D5 non passée : hors du segment (D2 pt 6 l'applicative au J28)

Segment [1787770800 ; 1788980400] = [2026-08-26T19:00:00+00:00 ; 2026-09-09T19:00:00+00:00] (J14p l.31), ex ante rule of ADR-0022 pt 5 (D4); Pyth removed from both strata, case (a): 0 ok readings out of 13,373 and out of 3,941 (l.38-39); skipped windows 1,027 and 1,819 (l.33).

J14p: quantity	calm	stress	J14p
n_s	13373	3941	l.209555; l.209575
K_s	50	8	l.209555; l.209575
P_more,s	0.0013938968610775433013870221315659599986142913741574	0.00035260783541832319883608669380651774613270860280165	l.209571; l.209591
guard (threshold 10)	18.614599673443475762938726299057869492601437469783	1.3891374858460684507352497516803467029217226329697	l.209572; l.209592
z_s	7.2684387263613165124682538101673599947301495076272	not published (guard §5.4)	l.209573; l.209593
z_bloc,s	2.6455172861617278377464936516549168389991781470697	not published (block guard, n_s < 7200)	l.209601; l.209605
rule value (hors décision)	REJETTE	NON ÉVALUABLE : strate non testée, hors décision (§1 bis.1 pt 2) [not evaluable: stratum not tested, outside the decision (§1 bis.1 pt 2)]	l.209620; l.209622

Lines of the rule (J14p l.209619 to l.209623), copied by extraction:

— règle SHOGEN-CRITERE-R1-1 (ADR-0028 §1 bis.1 pts 1-11 ; forme scellée) : valeur par strate ; comparaisons sur les Decimal publiées, non arrondies, au seuil 2.33, « ≥ » ; aucune p-valeur
« calme » : z_s = 7.2684387263613165124682538101673599947301495076272 ≥ 2,33 ; z_bloc = 2.6455172861617278377464936516549168389991781470697 ≥ 2,33 → REJETTE
« calme » : « le modèle d'indépendance du pool (k nominal_s = 11 flux du pool de la strate, bloc 1 ; k_eff mesuré ≤ 10 (borne supérieure), bloc 6) est rejeté dans la strate calme sur 13373 fenêtres, axes panne / staleness / hors-enveloppe (résidu : staleness fail-open : binance, kraken, bitfinex), tel qu'observé par cet instrument (hôte, DNS et réseau du harnais compris) ; aucune dépendance de paire n'est établie »
« stress » : z_s non publié (garde §5.4 : n·P_more·(1 − P_more) < 10) → NON ÉVALUABLE : strate non testée, hors décision (§1 bis.1 pt 2)
« R1 discrimine » (§1 bis.1 pt 6 ; déclencheur de D6 (vi) et D9) = VRAI : strate(s) qui rejettent : calme

The family counts m = 1 (l.209608); the pooled stratum is not published, the stress stratum being under the guard (l.209614). In block 6, the ten ASN probe records retained at the end of the segment (2026-09-09T18:49Z to 18:51Z) are all resolve_failed (J14p l.209777-209786): k_eff is known only as an upper bound, « k_eff ≤ 10 (BORNE SUPÉRIEURE) » [k_eff ≤ 10 (UPPER BOUND)] (l.209876); flag 1 true in stress only (l.209907); flag 2 not evaluable, « ni levé ni éteint » [neither raised nor off] (l.209908-209909), by reading (a) of pt 10. J14 is non-confirmatory and its publication is a decision of the investor (D4; decision 270). **This « R1 discrimine » VRAI is hors décision and does not change the verdict** (pts 1 and 9).

7.2 Second J14 (« seconde coupe » sensitivity)

Label (J14s l.1):

[ÉTIQUETTE] j14-second : sensibilité de la liste fermée (seconde coupe, décision 270), hors décision, non confirmatoire ; plage D5 non passée : hors du segment (D2 pt 6 l'applique au J28)

Segment [1787770800 ; 1788480060] = [2026-08-26T19:00:00+00:00 ; 2026-09-04T00:01:00+00:00] (J14s l.31), cut « ≤ 2026-09-04T00:00Z » of decision 270 (D4); Pyth removed from both strata, case (a) (l.38-39).

J14s: quantity	calm	stress	J14s
n_s	8121	1140	l.112112; l.112133
K_s	5	1	l.112112; l.112133
P_more,s	0.000014316253783038939038177418104725077784697432524987	0.000021469618207937163532218656623490189731153357463892	l.112128; l.112149
guard (threshold 10)	0.11626063253151037288958691789311493515448564399261	0.024474839280311532597570565581073686929332258857146	l.112129; l.112150
z_s	not published (guard \$5.4)	not published (guard \$5.4)	l.112130; l.112151
z_bloc,s	2.0943484839416630978298953875433710863142063001875	not published (block guard, n_s < 7200)	l.112159; l.112163
rule value (hors décision)	NON ÉVALUABLE : strate non testée, hors décision (§1 bis.1 pt 2)	NON ÉVALUABLE : strate non testée, hors décision (§1 bis.1 pt 2)	l.112178; l.112179

Both strata are under the \$5.4 guard, hence not tested: « R1 discrimine » = NON ÉVALUABLE, « aucune strate testée » [*no stratum tested*] (J14s l.112180); m = 0 (l.112166). In calm, z_bloc is printed (block guard met, n_s ≥ 7,200), but the value of the stratum remains NON ÉVALUABLE, the \$5.4 guard taking precedence (pt 5). Block 6: k_eff = 4 for nominal k = 10 (l.112437-112438), partition dated from 2026-09-03T23:56:54Z to 23:57:06Z (l.112439); flags 1 true in both strata (l.112452); flag 2 not evaluable (l.112453-112454). **Hors décision; does not change the verdict.**

7.3 Pooled stratum (exploratory, outside the family, hors décision)

On J28, stratified form of D2 pt 4, each stratum on its D1 pool (J28 l.435505-435510):

- z_pool = 33.435366780260737654814166344529336014557255168635 (l.435510);
- Σ_s (K_s - n_s·P_more,s) = 236.77931489141074698973370238916004307092589875631 and Σ_s n_s·P_more,s(1 - P_more,s) = 50.150507915819141795538676919575811501998150245996 (l.435509).

Reading: z_pool ≈ 33.4 ([calc] recounted at 33.4353667803 from the two printed sums). The pooled stratum is never among the inputs of the rule nor among those of flag 2 (pt 9; « strate poolée hors des entrées » [*pooled stratum outside the inputs*], l.435797). Its normal approximation is not measured (SHOGEN-POOLEE-BLOC-1, §11). It is published in neither of the two J14 renderings (strata under the guard, J14p l.209614, J14s l.112172). **Hors décision; does not change the verdict.**

7.4 « plage incluse » sensitivity (closed list, D2 pt 7)

[SENSIBILITÉ] section of J28 (l.435799-435820). Printed label: « variante “exclue” = principale (blocs 1-6) ; variante “incluse” = sensibilité — biaisée vers le haut par construction ; documente l'exclusion D5 ; pas un estimateur alternatif » [*variant ‘excluded’ = main (blocks 1-6); variant ‘included’ = sensitivity — biased upward by construction; documents the D5 exclusion; not an alternative estimator*] (l.435800, inner quotation marks rendered as “ ”); « motif de l'exclusion (harnais dégradé, [...]) : causalité non établie » [*reason for the exclusion (degraded harness, [...]): causality not established*] (l.435801). R1 alone per variant, neither L&M nor R2 (l.435802).

stratum	variant	n	K	P_more	z	flag 1	
calm	excluded (main)	24585	154	0.0013629504227977629135010363906355205959215005903986	20.829495417808173515660354538360738280133057485734	False	l.4
calm	included (sensitivity)	26294	408	0.0039561837661865178605964292161309130218132671860844	29.863015876556250088619968054838673686543339634366	False	l.4
stress	excluded (main)	11397	133	0.0014663989614904143004161900453685779659861375123928	28.466243694685921454102240881955034913016073716310	False	l.4
stress	included (sensitivity)	12306	133	0.0013272367906723694408097789913642102856351014485357	28.887094091051972755366700337792595909958417732478	False	l.4

- Difference in z (included — excluded): 9.033520458748076572959613516477935406410282148632 in calm (l.435809), 0.420850396366051301264459455837560996942344016168 in stress (l.435812).
- Losses by type removed by the range (SHOGEN-DP-JOURNAL-LOSS-1, l.435803-435806): those of \$2.4, plus, within the range, 73 grid windows without a marker in calm and 0 in stress, and no reading absent from the windows with a marker (D1 pool of the included variant).
- Coverage per weekend: \$2.5 (l.435813-435820).
- The third-party recomputation also returns the included variant of J28, under the entry label sensibilité « plage incluse » de la liste fermée (D2 pt 7), hors décision, biaisée vers le haut par construction [*“range included” sensitivity of the closed list (D2 pt 7), outside the decision, biased upward by construction*] (RT l.13924). There it prints z_bloc 2.5084623121105959997807452114411311466491022293836 in calm (RT l.14934) and 1.747895967777101321272619734145717678436022586316 in stress (RT l.15167), and, in flag 2 of the variant, “r1_discrimine”: “VRAI” (RT l.18045), “rejette”: [“calme”] (l.18047-18049) and the state eteint (l.17705), with the reason printed at l.18046 (copied below). [report finding] Package \$7 avoided a complete rendering of this variant, which would print « R1 discrimine » without a label (against pt 9); the JSON of the third-party recomputation carries this key for the variant, labelled at the entry level only. **Hors décision, biased upward by construction; does not change the verdict.**

“raison”: “« R1 discrimine » VRAI (strate(s) : calme) mais k_eff = 4 < k nominal du segment = 10 : recouvrement R2 mesuré explique au moins en partie la co-défaillance”,

7.5 L&M (block 4)

Difficulty function Θ over $N = 11$ feeds, D1 analysis pool (J28 I.435539); signed ϕ correlations per pair of feeds, complete co-anomaly matrix, « Cov<0 possible » (I.435544, I.435546).

L&M (block 4)	calm	stress	J28
Σm_i	1451	693	I.435541; I.435606
$\hat{E}(\Theta)$	0.0053654297705548468208626841939837668940780594227818	0.0055277704659120821268754935509344564359041853119242	I.435542; I.435607
$\hat{E}(\Theta^2)$	0.00058498345258564904690591084733854715550871743672239	0.0011262932031555353482176330294256064195521947561954	I.435543; I.435608
$Va\hat{r}(\Theta)$	0.00055619561596289281070470998663090781236412718685660	0.0010957369568317254707066076636531337727625164483380	I.435544; I.435609
ϕ defined / 55	55	55	I.435545; I.435610
pairs with a co-anomaly (n11 > 0)	55	50	I.435545; I.435610

The highest ϕ are, in calm, $\text{kraken} \times \text{bitfinex}$ and $\text{okx_ticker} \times \text{okx_index}$ (≈ 0.454 and ≈ 0.452 ; I.435563, I.435548); in stress, $\text{coingecko} \times \text{chainlink}$ (≈ 0.901 ; I.435614), $\text{kraken} \times \text{bitfinex}$ (≈ 0.763 ; I.435621), $\text{defillama} \times \text{chainlink}$ (≈ 0.691 ; I.435613) and $\text{coingecko} \times \text{defillama}$ (≈ 0.686 ; I.435612). Five ϕ are negative in stress, none in calm (I.435662-435666). These are descriptives: no per-pair significance (D3), no pair dependence established. **Hors décision; does not change the verdict.**

7.6 Exact tails

Under the \$5.4 guard, the renderings print the exact binomial tail hors décision (J14p I.209594; J14s I.112131, I.112152); no stratum of J28 is under the guard, and no tail is printed there. These values are not copied here (reading conventions). **Hors décision; they do not change the verdict.**

8. Checks of the execution

8.1 Seal

- Package** docs/adr-0028/PAQUET-PREREG-S2.md, commit 3be95be, sha256 4d2a8276316b1c66a08aabb15ff3b39be812978e93dafc01a4e627f20d0af528, written to the JOURNAL on 2026-10-03 at 01:02:51 UTC (JOURNAL I.304; docs/adr-0028/sceau/README.md). [calc] sha256 recounted equal on 2026-10-04.
- Timestamped **manifest** docs/adr-0028/sceau/PAQUET.sha256, sha256 519423510b21ecbebd895d6e225ab0748fb0fe5a2b257a36c9301e36bfaa3e9 (recounted equal).
- FreeTSA RFC 3161 token** paquet.tsr, sha256 9edb19b53379f370e291c5da01d4df3bc269b67a44feddb589d1211ec5ec6b0e, serial 0x08CFC8D5, **genTime 2026-10-03T01:04:10Z**, FreeTSA's clock (README of the seal; JOURNAL I.306). Replayed by the drafter on 2026-10-04: openssl ts -reply -in docs/adr-0028/sceau/paquet.tsr -text prints Time stamp: Oct 3 01:04:10 2026 GMT; bash scripts/sceau/verify.sh exits 0, with docs/adr-0028/PAQUET-PREREG-S2.md: OK (the manifest rechecks the bytes of the package) and Verification: OK twice (the token against the request, then against the bytes of the manifest and the FreeTSA chain). OpenSSL's warning "is not a CA cert" bears on tsa.crt, passed as signing certificate and not as authority.
- Withdrawal period** (A-7): 24 h from genTime, deadline 2026-10-04T01:04:10Z (README of the seal).
- Opening**: go of the investor, verbatim « go exécution » [*go execution*], recorded on 2026-10-04 at 01:06:36 UTC, after the deadline (JOURNAL I.310); route (a) (token), T0 = genTime (JOURNAL I.314).
- First seal, cited**: package 494d770d704dc7c342f0c6deec269e5642b3621bf451922f991e4ca1fb968097 (commit ddf8c54), manifest 680a95fdf50908cea797ae4fe9f0b42c43b5a99fcad755cb410e26fa6e794209, token serial 0x08CC76D7, genTime 2026-10-02T17:44:30Z (docs/adr-0028/sceau/premier-2026-10-02/README.md; [calc] sha256 of the manifest and genTime reread equal). It is cited because rule A-8 requires it: a seal replaced before the execution remains cited ("the first sha remains cited", ADR \$1 bis.8; package \$9). It was replaced before any execution (\$10.3) and no longer opens any execution (README of the first seal, addition dated 2026-10-03).
- What the seal does not attest**: the token attests the existence of the bytes of the manifest at the latest at genTime, nothing more, and nothing about earlier readings of the data (annexes D.1, D.3); it is checked under trust in FreeTSA, an individual operator, with no service-level agreement read; the OpenTimestamps durability anchor is not done (README of the seal; package \$12 pt 16).

8.2 Guards (annex D.4 b; package \$9)

- gardes-seules on the real journals at 01:11:19 UTC: « gardes levées » [*guards cleared*], exit 0; production launched at 01:11:27 UTC, process 943, author claude-opus-5-5, guarded head 27b0e303f196699959bddd2607db6a1d808ce0db, route (a) (JOURNAL I.314).
- Read-only replays by the drafter, on 2026-10-04: guard (1), the sha256 of the package appears in the JOURNAL of the guarded head, I.304 (git show 27b0e30: JOURNAL.md, sha alone displayed); guard (2), git diff --quiet f35a70c 27b0e30 -- s2-harness/shogen_s2 s2-harness/tools exits 0 (likewise against heads 81c5ad7 and 25406c8 of the branch during the drafting of the report), and f35a70c is an ancestor of 27b0e30; guard (4), the sha256 of s2-harness/tools/rendu_unique.py at commit f35a70c is 06d189cf84e7dca97bdfc0e1b695fc827a09473779505c7a8f1af1e7c26e9050, equal to the key sha256_script of the machine block (package I.216). Guard (4) is a guard against accidental editing, not a proof (package \$9).
- Guard (3): sha256 of the downloaded journals equal to the machine block and to the sums file (JOURNAL I.312; \$8.3).

8.3 Journals and outputs

- Sealed journals (machine block, package I.217-219; JOURNAL I.312): control.jsonl 351f51b2e4b7421b4ee286c27465cde239124d6edd70c0e550741d22f83366ff; journal.jsonl 98c5793ec460e3c009b7f743796800ae7259f71c65a4063a3c315e8eea595d74; raw.jsonl 39ffb13fb0e5939ff88285ce256fbd416b0665c20ee6394c512eed7d2175c15d; closing sums file 70910984076474987239d8c7a9da786acc95c38b375ad572d78afa297bf8caf4 (package I.220). Downloaded from a private storage without any display of content, sizes 41,676,139, 148,930,880 and 204,107,558 bytes equal to the expected ones, sha256 equal (JOURNAL I.312).
- Outputs: the seven sha256 of the conventions table, recounted by the drafter, are equal to those that the JOURNAL records (I.314) and to those of the runs of the record (ENR I.31-132).

8.4 Oracle record (D6 (viii))

ENR, schema shogen.oracle-record.v1 (I.138), role rendu (I.15), author claude-opus-5-5 (I.2), base f35a70c19ba8269f1f7e2bcd31775e4fc513da20 (I.3), ecrit 2026-10-04T01:11:29Z (I.4), variable SHOGEN_S2_CAMPAGNE_CONTROL at null (I.8), exit 0 (I.10), paquet.sha256 equal to the sha of the package (I.12), Python

3.11.15 (l.14), sceau.genTime 2026-10-03T01:04:10Z (l.136), served_from null (l.139), static_only false (l.140), tree.commit 27b0e303f196699959bddc2607db6a1d808ce0db extracted by git archive, 436 files hashed (l.142; [calc] count of the entries). Six runs, in the order suite, j14-principal, j14-second, j28, recalcul-tiers, raw, each with exit 0 and without any test launched with the variable (tests_avec_variable empty) (l.31-132). Reread by oracle_record.py --verifier ... --role rendu --commit 27b0e303...: « conforme » [conformant] (JOURNAL l.314).

8.5 s2-harness suite

Ran 398 tests in 62.358s (SUITE l.621) and 0K (skipped=2) (l.623). The two skipped tests are the count tests that would read a sealed copy through SHOGEN_S2_CAMPAGNE_CONTROL (SUITE l.207, l.209): the variable was not set (ENR l.8; annex D.4 a).

8.6 recompute_* oracle (run recalcul-tiers): what it establishes and what it does not establish

- Object:** the JSON of the recompute_* entry points of the recomputation path (ADR-0028 D6 (i); third-party oracle of ADR-0003) for the main J14, the second J14, J28 and the included variant of J28 (RT l.3, l.5048, l.9462, l.13897), each with its label (RT l.23, l.5060, l.9489, l.13924); no warning from the reader ("avertissements": [], RT l.2).
- What it establishes [calc]:** the 126 compared values (n, K, $\hat{P}_{more}$, guard, z, $\hat{y}_0$, $\hat{\sigma}^2_{bloc}$, cv, FIV and its factors, z_bloc, runs, decomposition of K, state of flag 2 and « R1 discrimine », for J28, the main J14 and the second J14) are equal, string for string, to those of the renderings (comparer_recalcul.py, §8.9). In this execution, the recomputation path and the report path therefore return the same values from the journal alone.
- What it does not establish:** an independent implementation. The two paths call the same frozen modules (r1, lm, r2, records), under the same Python, on the same host, in the same execution: it is a consistency check between two entry points, not a recomputation by third-party oracle. The independent oracle in exact arithmetic, written from the sealed text, was compared with r1 on 585 synthetic journals before the seal, with 0 differences (annex B.43), and not on the real journals. Nor is it the composition test SHOGEN-S2-TUYAU-MONARK-1: "The recompute_* oracle, replayed in the single execution, is an internal check: it is not the composition test" (ADR §3). It says nothing about the fidelity of the journals to the sources (A(history-integrity), doc 08).

8.7 Verdict of the raw journal

RAW l.1, copied by extraction:

```
verdict raw.jsonl (records.verifier_raw ; SHOGEN-RAW-FIN-1) : refus – lecture(s) de raw.jsonl absente(s) de journal.jsonl : [(1790275920, 'binance', Decimal('1790275970.0000675')), (1790277780, 'binance', Decimal('1790277830.0003345')), (1790283600, 'binance', Decimal('1790283650.0009525')), (1790283840, 'okx_ticker', Decimal('1790283890.0004282')), (1790285940, 'binance', Decimal('1790285990.0001917'))] – refus (SHOGEN-RAW-LECTEUR-1)
```

- The raw-journal verifier refuses (SHOGEN-RAW-LECTEUR-1): five readings of raw.jsonl are absent from journal.jsonl, four from binance and one from okx_ticker. [calc] Their window_start values are 2026-09-24T18:52Z, 19:23Z, 21:00Z, 21:04Z and 21:39Z: all within the D5 range [1790273880 ; 1790435280] (contrôles_r1.py), hence outside n, K and $\hat{P}_{more}$ of J28 (§2.4), and outside the two J14 segments, which end on 2026-09-09 and 2026-09-04.
- This verdict does not close the execution, by a construction written before the seal (package §12 pt 1, SHOGEN-RAW-FIN-1): "The single execution prints and records this verdict without the verdict closing it (named run that exits 0 whatever the verdict [...]; the list of guards of D.4 b is closed)". The raw run exited 0 (ENR l.126-127).

8.8 Duration

Production launched at 01:11:27 UTC, ended at 01:33:47 UTC, exit 0 (JOURNAL l.314), i.e. 22 min 20 s [calc]; the suite ran in 62.358 s (SUITE l.621). A single execution, no failed attempt (JOURNAL l.314).

8.9 Recounts by the drafter [calc]

Scripts of the drafter, Python standard library, filed in the project repository with this report. The first three were initially run under a mutation that must make them fail, then normally (G1 journal); calc_divers.py has no mutant mode: it performs elementary operations on copied values, cited line by line in its code.

script	object	result	sha256 of the script; of the output
contrôles_r1.py	from the printed integers (anomalies per feed, n, K, s): exact $\hat{P}_0$, $\hat{P}_1$, $\hat{P}_{more}$; z_s, $\hat{y}_0$, z_bloc (printed $\hat{\sigma}^2$), FIV and factors, cv, EMD_s, censoring bounds, z difference of the sensitivity, z_pool; rule value; integer identities; raw verdict against the D5 range	maximum relative difference from the printed values $4.21 \cdot 10^{-50}$ (last digit of precision 50); recounted rule value equal to the printed one in both strata	11779f798b948b155462c7b66460009a0a2c991bb27653c2ad344c6c83744c4f; 0469a499233d66cd64031c7cf21e82bf01615658e3f14cc25d8221afe5581e3e
comparer_recalcul.py	JSON of the third-party recomputation against blocks 3 and 6 of the renderings	126 values out of 126 equal as strings	4e61cea803c902054e6fcbd2cc6a29f7c71ac1a2f8bfd824aefaa851ebef7585; 8e6731f9f3c6f0a73ad2fb7938e66c215dafa1d0c13934f4d6f804d6dd57459e
d3_lift.py	lift and ϕ identity of D3 (§6)	exact identity on 110 tables out of 110; difference from printed $\phi \leq 2.38 \cdot 10^{-50}$	8f91474cdded5694e733334fadf225b110d3378ddb560f1bee4066eca80f9381; 77ae69ec8666065d0e7ec3a13588f982d87de7bf1789ffe3f91813ea028dc73f
calc_divers.py	roots, duration, sums, τ ratios, z_pool, guards of the J14 segments, sums of the anomalies per type	values cited [calc] in the text	4d210cb65fb56fc5c0ef00ec31480bbe96426746191c5efa38a9e2def99a1c98; de7bd825bba4e71aeac9437dfafa26eb3101d00c3d1ea0b992b103c33f4c52af

$\hat{\sigma}^2_{bloc,s}$ cannot be recomputed without the l_t series, hence without reading the journal: it is taken as printed, and the recount of z_bloc,s checks only the division by its root. The tables of this report were extracted from the renderings by scripts (extraire_tables.py, extraire_d5.py, extraire_hd.py, table_lift.py), without copying by hand.

9. Limitations

9.1 Limitations written with the rule (package §10.2, pts 2, 7 and 11)

1. Serial dependence of range $\geq l$ not corrected: $\hat{\sigma}^2_{\text{bloc},s}$ is biased downward there (pt 11).
2. Even under a range $< l$, bias of the triangular-weight estimator of the order of $-\ell^{-1} \sum_k |k| \cdot R(k)$: under positive autocorrelations, $\hat{\sigma}^2_{\text{bloc},s}$ underestimates and $z_{\text{bloc},s}$ overestimates (pt 11, SHOGEN-BARTLETT-BIAIS-1).
3. z_s and $z_{\text{bloc},s}$ are conservative through the plug-in of $\hat{P}_{\text{more}}$; a non-rejection is thereby less probative (pt 11; SHOGEN-R1-PLUGIN-1).
4. Reduced, even negative, power against a common mode that affects all the sources, as soon as $\sum_i (\partial P_{\text{more}} / \partial p_i)(1 - p_i) > 1 - P_{\text{more}}$ (pt 11; SHOGEN-R1-PLUGIN-1).
5. Normal approximation at the guard; near the guard, with iid windows, the level of z_s alone is measured above 0.01 on synthetic data, whereas the rule holds (pt 11, SHOGEN-GARDE-NIVEAU-ZSEUL-1; §3.4).
6. Asymptotic level, not demonstrated ≤ 0.01 in finite samples; the bound 0.02 holds under the joint null model (pt 7).
7. Under the §5.4 guard, declared departure from doc 10 §5.4: the stratum is not tested, its exact tail is hors décision (pt 2).
8. “No author of the rule has seen z , K , $\hat{P}_{\text{more}}$ or ϕ (annex D.3)” (pt 11), under the limitations of the D.1 inventory and of the measurements of annex B.37 (§10).

9.2 Limitations and procedures written in the package (§12, points 1 to 20)

1. SHOGEN-RAW-FIN-1: the raw-journal oracle remains strict; its verdict is printed without closing the execution (§8.7).
2. SHOGEN-ENREG-G1-1: the recorder extracts only a commit; a G1 record of uncommitted work cannot be produced.
3. SHOGEN-ENREG-DELAI-CHAMP-1: the timeout (3,600 s per command) is not a field of the record; it kills only the direct child.
4. SHOGEN-CENSURE-VIVANT-PORTE-1: « harnais vivant » means the same start before and after the skipped window; a sleep during a start is counted « vivant » *[alive]*.
5. SHOGEN-RENDU-TABLE-REELLE-1: the real output table never ran end to end on a fixture; the single execution is its first complete execution.
6. SHOGEN-RENDU-CLI-REPORT-1: `python -m shogen_s2.report` renders journals without a guard; procedure: no rendering of the sealed journals outside `rendu_unique.py`.
7. SHOGEN-RENDU-JETON-MAIN-1: the token `SHOGEN_RENDU_PRODUCTION` blocks the accidental call of `--produire`, not the deliberate call.
8. SHOGEN-RENDU-ORC-OCTETS-1: guard (4) hashes the files on disk, not the bytes loaded in memory.
9. SHOGEN-RENDU-RENAME-FENETRE-1: under POSIX, a window remains between the exclusive creation of the target and the renaming.
10. SHOGEN-GO-PICKAXE-FUSION-1: `git log -G` does not read merge commits; effect in refusal only.
11. SHOGEN-AXES-SIGMA-NUL-1: `axes_evaluables` marks « staleness » evaluable for a class σ of None; without effect in production.
12. SHOGEN-PAQUET-ERRATUM-FISHER-1: the « (Fisher) » of doc 10 designates the transformation $z' = \text{artanh}(r)$; the interval “ $\approx \pm 0.11$ ” holds on the z' scale; $SE = 1/\sqrt{N-3}$ assumes independent pairs; $N_{\text{min}} = 300$ is a design choice.
13. SHOGEN-TAU-REDERIV-1: τ is not re-derived; the committed τ serves all the confirmatory tests (§5.1).
14. SHOGEN-CENSURE-INFO-1: informative censoring is not modelled; the verdict is not identified under arbitrary censoring of the skipped windows (§5.3).
15. SHOGEN-SEG-DEMARRAGE-1: a record belongs to the segment by its own timestamp; a segment without a retained `asn_attribution` makes `k_eff` not evaluable.
16. Private seal: with anchor (1), the token attests the existence of the bytes at the latest at `genTime`, nothing about earlier readings, under trust in FreeTSA; in the cloud session, the commits are signed by the key of the session and no offline bundle is made (§8.1).
17. Harness clock (T-17): the assignment of windows to strata and segments rests on the host's clock, `A(harness-clock)`; the range of the median offsets of the clock check does not enter block 1 (§2.7).
18. SHOGEN-COLLECT-PREVWS-1: item not qualifiable in the cloud session, outside the rule; requalification after S2.
19. SHOGEN-ATTEST-ADVISOR-1: attestations of advisors missing; class R is excluded by the D.1 inventory, not by attestation.
20. Values of the machine block: the sha256 of the journals were only on the local workstation; the package validator accepted their text and format, not the values. At the execution, the downloaded journals are equal to them (§8.3).

9.3 Limitations observed from the execution

1. [report finding] The verdict is decided entirely on the standard-error floor: $z_s \approx 20.8$ and ≈ 28.5 , but $FIV_s \approx 115$ and ≈ 266 bring $z_{\text{bloc},s}$ back below 2.33 (§3.1). The rule does not separate co-failure of the sources from serial dependence of the windows: this is the sealed statement of the discordance.
2. [report finding] The counted anomalies are almost all outages: in calm, 1,443 of the 1,451 anomalies (4 of staleness, 4 out-of-envelope); in stress, 691 of the 693 (2 of staleness, none out-of-envelope) [`calc`, sums of block 3, equal to $\sum m_i$ of block 4]. K is formed, to within one window, of windows with at least two `panne_transport` readings (§5.2). These counts describe; they identify no cause.
3. [report finding] The censoring is large relative to n : 7,795 grid windows skipped outside the D5 range, of which 7,628 without a cause attributed by the journal, for 35,982 analysed windows, and 4,093 starts of the harness (§2.6); with $\hat{\sigma}_{\text{bloc},s}$, the non-outer bounds range from ≈ 1.82 to ≈ 93.9 in calm and from ≈ 1.70 to ≈ 33.1 in stress (§5.3). H_{perte} is an assumption, not tested.
4. [report finding] The R2 partition is a snapshot: one probe record per host, the last one, taken between 2026-09-28T01:17:55Z and 01:18:02Z (J28 I.435781). It does not describe the ASN axis over the 33 days. On the main J14, the ten probe records retained at the end of the segment failed and `k_eff` there is only an upper bound (§7.1).
5. [report finding] A single observer: one collection host (Windows traces in the renderings; the investor's workstation, JOURNAL I.324), one DNS resolver (Cloudflare DoH) per probe record; the multi-resolver and multi-vantage probe is « DUE » and residual 5 is « à son MAXIMUM, publié » *[at its MAXIMUM, published]* (J28 I.435772, I.435774). The anomalies of R1 and the ASN axis of R2 depend on the same observer.
6. [report finding] The staleness axis is not evaluable for binance, kraken and bitfinex, which carry no timestamp (J28 I.435464): it covers 8 of the 11 feeds of the pool.
7. [report finding] Uneven coverage of the stress stratum: the weekend of 2026-08-29 is covered over only 19 h of the 48 h, and that of 2026-09-26 over 32 h 51 min in the main variant, because of the D5 range (§2.5).
8. [report finding] No price data was received from Pyth over the whole campaign: access to the queried service was refused (HTTP 401) from the first day (ADR I.25), and nothing shows an outage of Pyth; the pool loses one of the two oracle-class feeds, and chainlink is observed only through a public RPC (§4.4).
9. [report finding] The label of block 5 (d) of J28, « aucun cluster à ≥ 2 flux (1) » *[no cluster with ≥ 2 feeds (1)]*, reads poorly next to the partition of block 6; the main D3 estimator between clusters has no object on J28 (§4.2).
10. [report finding] The JSON of the third-party recomputation carries “`r1_discrimine`”: “VRAI” for the « plage incluse » variant, labelled only at the entry level (§7.4); a reading of the line alone would isolate it from its label.
11. [report finding] Two hors décision outputs cross the threshold where J28 does not cross it: the main J14 REJETTE in calm ($z_{\text{bloc}} \approx 2.65$) and the included variant gives $z_{\text{bloc}} \approx 2.51$ in calm (§7.1, §7.4). Pt 9 excludes them from the decision; this report draws no reading from them.
12. [report finding] The third-party recomputation is not an independent implementation (§8.6); $\hat{\sigma}^2_{\text{bloc},s}$ is recounted by no distinct code on the real journals.
13. [report finding] The observed τ exceeds τ_{classe} at the maximum for two classes (§5.1): descriptive finding, without re-tune; the re-derivation is PX-Shogen-13 (§11).
14. [report finding] The byte-for-byte equality of a replay of the renderings on another host or another Python version is not measured (§12).

10. Declared deviations and exposures

10.1 Reading D.1 No. 16: exposure of the orchestrator of the cloud session

- **What is declared** (annex D, D.1 No. 16, addition dated 2026-10-02 17:5x UTC, declared deviation under D2 pt 8): while measuring the item MONARK-S2-M009A-EXPOSITION-1, the orchestrator of the cloud session displayed the subjects of the commits of a private repository of the MONARK project that touch documents forbidden to fresh drafters and validators (annex D); the subject of one of these commits carries a rate estimated on the S2 traces, whose value is copied nowhere; the content of the files was not downloaded (clone without blobs). Date: 2026-10-02 around 17:57Z; class R presumed (same measurement as reading No. 2, MONARK reimplementation of `r1.classify_ecart` on a snapshot of 18/09; exact nature of the rate not established without reading).
- **Position in the chronology**: after the first seal (genTime 2026-10-02T17:44:30Z) and before the second (2026-10-03T01:04:10Z). The revised package lists it: annex D “counts 16 readings in D.1 (No. 16, exposure of the orchestrator of the cloud session, is dated 2026-10-02, after the first seal and before this one)” (package I.8, (v)). The text of the rule (§10.2) was already sealed by the first package and it is identical byte for byte in the second (package I.8; JOURNAL I.304). The single execution was conducted by “the orchestrator of the cloud session” (JOURNAL I.310); it is the mechanical execution of a sealed script, under the guards of §8.
- The item SHOGEN-MONARK-SUJET-COMMIT-1, which measured the sessions that had displayed this commit subject, is closed (annex B.38).

10.2 Closure of MONARK-S2-M009A-EXPOSITION-1 (annex B.38)

Annex B.38 asks that this result be “taken up in the report of the execution”; it is taken up here, in substance. The item is closed after the sealing (trigger “before sealing” passed, departure recorded in package §8 and in the JOURNAL). The readers of the figure of reading D.1 No. 2 are the MONARK sessions of the M009 work packages, the orchestrator of the local workstation and its sub-agents, and a mapping sub-agent (commit subject only, not transmitted); no author of rule SHOGEN-CRITERE-R1-1 and no decision-maker, under the limitations of annex B.37 (search by exact tokens, window of 300 characters, text files); plus the cloud orchestrator after the first sealing (D.1 No. 16, §10.1). Attestation of the investor, decision-maker of the rule, to the question of whether the investor saw this figure: “no” (annex D.3 (i)). The revised package notes that this closure lifts the limitation written in its §8 (package I.8, (v)), whose text has remained unchanged.

10.3 Replacement of the first seal (A-8; CORR work package)

- First package: commit ddf8c54, sha256 494d770d704dc7c342f0c6deec269e5642b3621bf451922f991e4ca1fb968097, FreeTSA token of genTime 2026-10-02T17:44:30Z; replaced before any execution; it remains cited and no longer opens any execution (package I.8; §8.1).
- Reason: two class A findings of the catch-up G2 reviews, each capable of leaving the single execution without any output, without producing a false value (annex B.42, A-1: a frozenset of pairs of exact copies that the third-party recomputation did not serialize; annex B.43, A-1: a non-finite ok price that raised a decimal exception).
- Decision of the investor of 2026-10-02: “Correct and reseal”. CORR work package, commits 02f9c00, 4002239, 35cd2e2, 4c831b8, e4bc2f1, a5a9de9, f35a70c, suite 398 OK, rule and pins unchanged at each commit (annex A, CORR line; annex B.44).
- Effect on the sealed text: sections 1 to 12 unchanged, section 10.2 identical byte for byte; no value, no threshold, no inequality of the rule changes; new machine block (analysis commit f35a70c) (package I.8; JOURNAL I.304).
- In this execution, neither of the two corrected cases arose: the list of exact copies is empty in the four entries of the third-party recomputation (RT I.1520, I.6557, I.10991, I.15421), and no warning from the reader is printed (RT I.2), whereas clarification (i) of the package provides for one, counted per feed, for any non-finite price.

10.4 Precedence D-4 (SHOGEN-D4-PRECEDENCE-RAPPORT-1, annex B.45)

The D-4 precedence (FAUX case with `k_eff` not evaluable, returned NON ÉVALUABLE for flag 2) is not written in the package; it is ratified (annex B.18) and must be cited in the report if the case arises. **The case does not arise on J28**: « R1 discrimine » is FAUX and `k_eff` = 4 is evaluable (J28 I.435780); flag 2 is off by pt 10 itself (I.435795). Nor does it arise in the hors décision outputs: on the main J14, « R1 discrimine » is VRAI with a `k_eff` upper bound, hence a flag not evaluable by reading (a) of pt 10 (§7.1); on the second J14, « R1 discrimine » is NON ÉVALUABLE, hence a flag not evaluable by pt 10 (§7.2).

10.5 Other deviations

No other deviation is known to the orchestrator at the date of the brief of this report (2026-10-04). The JOURNAL records a single execution, with no failed attempt and no second execution (I.314). The refusal « run_params divergents » [*diverging run_params*] that item SHOGEN-R2-RUNPARAMS-CONCORDANCE-1 feared did not occur: the execution wrote its seven outputs (annex B.45; §8.4). The decisions of the investor recorded after the execution (JOURNAL I.322; §1) are those that D6 (vi) and D9 reserve to the investor after J28, plus an S2-bis work package; to the drafter’s knowledge, none changes a decision prior to the seal (ADR §1 bis.8 would make it a declared deviation).

11. Analyses added after the pre-registration: not done at the date of the report

Each will carry the mandatory label « ajoutée après le pré-enregistrement, hors décision » [*added after the pre-registration, outside the decision*]; none is computed here, none enters the rule or can change the verdict (package §7, §11).

item	what it will measure	source
SHOGEN-FLUX-QUASI-MORT-1	the sealed rule recomputed per stratum after removal of the “quasi-dead” feeds, $2 \cdot \text{ok}(f, s) < n_s$, threshold fixed in advance by an author without exposure to the presence rates; construction constraint SHOGEN-QUASI-MORT-PREDICAT-1: count « ok » by the predicate of <code>r1.analysis_pools</code> on the list of <code>r1.parse_journal</code>	docs/adr-0028/AVIS-SEUIL-FLUX-QUASI-MORT.md §1; annex B.39, B.44
SHOGEN-HOST-DEGRADED-2	the sensitivity “windows with a degraded-host diagnostic removed”, criterion defined on the diagnostic records of the harness alone (<code>asn_attribution.resolve_failed</code> , <code>clock_check</code>), never on a source status	annex B.6; annex D.5
SHOGEN-CENSURE-INFO-2	the outer bounds of <code>z</code> under arbitrary censoring of the skipped windows (arbitrary imputations, Manski-type bounds), with the reading “identified” or “not identified under arbitrary censoring”	annex B.6 (P-08); annex D.5
SHOGEN-DEP-FENETRES-2	the serial dependence of range $\geq l$ (fixed-b asymptotics, linearized version of Künsch, event statistic), discharge of $A(\text{window-dependence})$; priority before G10 if the maximal run is $\geq l$, which is not the case on J28 (6 and 2, §5.4)	annex B.6
SHOGEN-R1-PLUGIN-1	the limitations of the transposed <code>z</code> : variance of the influence function (conservativeness of the plug-in), test of the complete distribution of <code>m_t</code> against its Poisson-binomial distribution (absorption of a broad common mode)	annex B.6
SHOGEN-CONTENU-DEP-1	the serial dependence on the content axis (effective $N < N$ for $N_{\min} = 300$ and the Fisher SE), by a block variance of $\hat{p}$	annex B.6; package §12 pt 12
SHOGEN-POOLEE-BLOC-1	a block standard-error floor for the pooled stratum and the level of <code>z_pool</code> under the null model, by simulation	annex B.13, B.18

item	what it will measure	source
PX-Shogen-13	the re-derivation of τ after the rendering, never applied to the confirmatory z ; reservation: observed τ and calibration P99 do not bear on the same population	annex D.5; package §12 pt 13; annex B.45

11.1 Addition dated 2026-10-04: analyses done after the pre-registration (POST-PREREG work package), hors décision

Each output carries on its first line: « ajoutée après le pré-enregistrement, hors décision ; ne change pas le verdict de la règle scellée (« R1 discrimine » = FAUX, docs/11 §3) » [added after the pre-registration, outside the decision; does not change the verdict of the sealed rule (“R1 discriminates” = false, docs/11 §3)]. None enters the rule or replaces a value of a rendering; the verdict remains that of §3. Code: scripts/post-s2/, which reuses the readers of the harness without modifying s2-harness; outputs: docs/adr-0028/execution/post-prereg-2026-10-04/ (abbreviated PP/ below; sha256 in SHA256SUMS); G1 journal of the worker (docs/G1-lot-POST-PREREG.md), where the parameters are written on 2026-10-04 at 04:18:36 UTC, before any execution on the journals. Exposure declared at the time of fixing them: block 3 of the J28 rendering and this report. The threshold of FLUX-QUASI-MORT-1 comes from the advice of an author without exposure (annex B.39); that of FLUX-DEVIANT-1 ($\hat{p}_f > 1/2$) is the value of variant V2 of the same advice, applied by the worker. The constructions come from annex B, written before the execution, except the form of the criterion of HOST-DEGRADED-2 (ASN probe of the start, last marker, c_lock_check outside the criterion), a choice of the worker not written in the annex. These choices of the worker are fixed before the execution, after the exposure declared above, and named in the G1 journal. Consistency check: each output recounts n , K and $\hat{P}_{\text{more}}$ per stratum and finds them equal, string for string, to block 3 of J28 (PP/*.txt l.7-8). Roundings “ $\approx$ ” to 3 significant figures; complete values at the lines cited.

item	what is computed	result	output
SHOGEN-FLUX-QUASI-MORT-1 (with QUASI-MORT-PREDICAT-1 and POOL-MIN-1)	sealed rule recomputed per stratum without the feeds with $2\text{-ok}(f, s) < n_s$, ok counted by the predicate of r1.analysis_pools	no feed removed (minimum ok rate ≈ 0.981 in calm, bitstamp; ≈ 0.980 in stress, defillama [calc]); recomputed rule identical to the sealed rule; recomputed « R1 discrimine » = FAUX	PP/quasi-mort.txt l.10-16
SHOGEN-FLUX-DEVIANT-1 (exploratory, conditioned on $\hat{p}_f$)	same recomputation without the feeds with $\hat{p}_f > 1/2$	no feed removed (maximum $\hat{p}_f \approx 0.0193$ in calm, ≈ 0.0198 in stress [calc]); identical; FAUX	PP/deviant.txt l.10-16
SHOGEN-POOLEE-BLOC-1 (a)	$z_{\text{pool}, \text{bloc}} = \sum_s (K_s - n_s \cdot \hat{P}_{\text{more}, s}) / \sqrt{\sum_s \hat{\sigma}^2_{\text{bloc}, s}}$	≈ 2.60 (binomial $z_{\text{pool}} \approx 33.4$); exploratory, outside the family; level not measured (POOLEE-BLOC-1 (b), DETTES-SIM work package)	PP/poolee-bloc.txt l.12-13
SHOGEN-HOST-DEGRADED-2	rule recomputed without the windows of the starts whose ASN probe carries a resolve_failed	631 starts out of 4,172; windows removed 3,341 in calm and 720 in stress; K goes from 154 to 70 and from 133 to 59; calm: $z_s \approx 18.1$, $z_{\text{bloc}} \approx 1.80$, NE REJETTE PAS (discordance), $\text{EMD}_s \approx 104$; stress: guard $\approx 6.03 < 10$, NON ÉVALUABLE; FAUX	PP/hote-degrade.txt l.10-18
SHOGEN-HORLOGE-ETENDUE-1	median offset of the 3,610 retained c_lock_check	none non-evaluable; minimum ≈ -52.2 s, median ≈ 1.42 s, maximum ≈ 96.3 s, range ≈ 148 s	PP/horloge-etendue.txt l.10-11
SHOGEN-CENSURE-INFO-2	outer bounds of z_s under arbitrary censoring of the skipped windows; rule value under two witness imputations	calm: $z_s \in [\approx -169; \approx 121]$; stress: $[\approx -88.6; \approx 81.5]$, bounds attained; all skipped windows clean: FAUX; 19 skipped windows in calm (42 in stress) imputed with two anomalies: REJETTE, hence « R1 discrimine » VRAI under this imputation; reading: not identified under arbitrary censoring	PP/censure.txt l.10-21
SHOGEN-SIGMA-BLOC-INDEP-1	classification, I_t series and $\hat{\sigma}^2_{\text{bloc}}$ by a code independent of r1	0 disagreements out of 270,435 cells in calm and 125,367 in stress; $\hat{\gamma}_0$ and $\hat{\sigma}^2_{\text{bloc}}$ equal, string for string, to block 3	PP/sigma-indep.txt l.9-13
SHOGEN-DEP-FENETRES-2 (b), which is SHOGEN-R1-PLUGIN-1 (a)	block variance of the influence function of $K/n - P_{\text{more}}(\hat{p})$ (Künsch 1989, Ex. 2.2 and (2.14))	$\hat{\sigma}^2_{\text{IF}, \text{bloc}} \approx 0.672 \cdot \hat{\sigma}^2_{\text{bloc}}$ in calm, ≈ 0.677 in stress; $z_{\text{IF}, \text{bloc}} \approx 2.37$ in calm, ≈ 2.12 in stress; exploratory, level not measured	PP/influence.txt l.10-11
SHOGEN-DEP-FENETRES-2 (c)	runs of I_t taken as units, iid null distribution at $\hat{P}_{\text{more}}$	$R = 119$ against $E[R] \approx 33.5$ in calm ($z_R \approx 14.8$); 130 against ≈ 16.7 in stress ($z_R \approx 27.8$)	PP/influence.txt l.13-14
SHOGEN-R1-PLUGIN-1 (b), descriptive	counts of m_t against the Poisson-binomial distribution of $\hat{p}$	calm: $m = 1$ observed 1,001 against $\approx 1,380$ expected, $m \geq 3$ observed 49 against ≈ 0.424 [calc]; stress: $m = 1$, 214 against ≈ 659 , $m \geq 3$, 118 against ≈ 0.224 [calc]; no test	PP/influence.txt l.17-40
SHOGEN-CONTENU-DEP-1	effective size N_{eff} of $\hat{p}$ (block variance of its influence function) against the Fisher SE	$\hat{p}$ equal to block 5 for the 55 pairs; median $N_{\text{eff}}/N \approx 0.0535$ (p_{raw}) and ≈ 0.0556 (p_{resid}); $N_{\text{eff}} < N_{\text{min}} = 300$ for 3 pairs (p_{raw}) and 2 (p_{resid})	PP/contenu.txt l.10-66

What these analyses allow one to say. The verdict of §3 depends on no almost-dead feed and on no feed with an anomaly rate above 1/2: there are none on J28. $\hat{\sigma}^2_{\text{bloc}, s}$, on which the discordance depends, is reproduced by a code distinct from r1 on the real journals, common readers aside (second half of limitation 12 of §9.3). The windows of K are concentrated in the starts whose ASN probe of the harness carries a resolve_failed: 84 of the 154 windows of K in calm and 74 of the 133 in stress fall there, for 13.6% and 6.3% of the windows [calc]. The FAUX verdict is not identified under arbitrary censoring of the skipped windows: it rests on the assumption H_{perte} (doc 08, A(loss-non-informative)); an imputation of 19 skipped windows in calm, or of 42 in stress, gives « R1 discrimine » VRAI. The estimated effective size of the content series is of the order of 5% of N at the median and remains below N for the 55 pairs, on p_{raw} as on p_{resid} : for each, the Fisher SE (package §12 pt 12), which assumes independent pairs, is smaller than the block SE (ratio from 1.50 to 15.0 [calc]).

What they do not allow one to say. None identifies a cause: the concentration of K in the degraded starts is compatible with a common mode of the observer, without establishing it. Two exploratory statistics with a block standard error cross 2.33 where $z_{\text{bloc}, s}$ does not: $z_{\text{pool}, \text{bloc}} \approx 2.60$ and $z_{\text{IF}, \text{bloc}} \approx 2.37$ in calm; none is in the rule or in the family, their level in finite samples is not measured, and they do not change the verdict (same status as the hors décision outputs of §9.3 pt 11). The runs taken as units keep an excess under an iid null distribution that ignores the persistence specific to each source; this is not a test of the dependence of range $\geq l$. The distribution of m_t is described, not tested. Nothing is said about the independence of the sources (doc 09).

Remaining. Not done, with their reason: the fixed-b form of DEP-FENETRES-2 (a) (Kiefer-Vogelsang, P-05, not filed); a tolerance and a null distribution of the event statistic (c); the test of the distribution of m_t of R1-PLUGIN-1 (b) (statistic and level to be sourced); an outer bound of z_{bloc} under arbitrary censoring (CENSURE-INFO-2; “Manski-type” attribution [inferred: P-08 not filed]); the level of $z_{\text{IF}, \text{bloc}}$ and of $z_{\text{pool}, \text{bloc}}$ (synthetic simulation). The c_lock_check does not enter the criterion of HOST-DEGRADED-2: its degradation signals depend on the sources that answer the probe; departure from the wording of the item, declared in the G1 journal.

12. Reproduce

- **Inputs:** the three sealed journals, designated by their sha256 (machine block, package l.217-219; §8.3), and the closing sums file (package l.220). Their format is specified by reference (package §1: records.py and journal.py at the collection commit ed479c5, doc 10 §6).

- **Code:** analysis commit f35a70c19ba8269f1f7e2bcd31775e4fc513da20 (s2-harness/shogen_s2, s2-harness/tools); single-execution script s2-harness/tools/rendu_unique.py, sha256 06d189cf84e7dca97bdfc0e1b695fc827a09473779505c7a8f1af1e7c26e9050 (package l.215-216); Python 3.11.15 at the time of the execution (ENR l.14).
- **Seal:** docs/adr-0028/sceau/ (manifest, request, token, FreeTSA chain); offline check by bash scripts/sceau/verify.sh (§8.1).
- **Replay:** with the journals and the commit, the recompute_* entry points recompute the blocks from the journal alone (doc 10 §6; ADR-0003); their values are compared, string for string, with those of the filed renderings (§8.6, §8.9). A new production by rendu_unique.py would be a second execution, a declared deviation (annex D.4 b).
- **This report:** each value in it is cited by file and line; the scripts of the drafter and their outputs (§8.9) replay its computations from the renderings alone, without the journals.
- **What is not public:** the project repository is private (package §12 pt 16); the sealed journals are not published, and their publication is a precondition for the composition test SHOGEN-S2-TUYAU-MONARK-1 on the MONARK side (ADR §3; §4 pt 4). The sealed package, the seal, the renderings, the records and the scripts are in the same private repository and are not published: documents provided on request, under agreement. Checking the seal requires the sealed package, which is provided together with the seal on request: a third party who is given the sealed package, the seal and the renderings without the journals can check the seal and the arithmetic of the renderings, not recompute the renderings.

Addition dated 2026-10-04 06:52:35 UTC (G2 review of the DETTES-B1 work package, C-3; points “Code” and “Replay” above): from commit 0221a74 (DETTES-B1 work package, B1-1; rendu_unique.py from 0789d96, B1-4), s2-harness/shogen_s2 and s2-harness/tools differ from the analysis commit f35a70c (sha256 of rendu_unique.py at the head 5ee95dbf... ≠ sha256_script 06d189cf...): guards (2) and (4) refuse at the head; any replay (recompute_*) or re-execution of the S2 rendering is done on an extraction of f35a70c; at the head, the outputs differ by construction (block 5 (d), k_eff note, key etiquette of flag 2 of j28-incluse, ASN divergences of partial probe records).

Addition dated 2026-10-04 09:36:41 UTC (closing cp-2, C-7; limitation to be read with §9.3): an ASN change visible on a single base of a partial probe record is not searched for by the S2 renderings (semantics of the CORR work package: complete probe records only); the effect of the later fix SHOGEN-ASN-DIVERGENCE-PARTIELLE-1 on the sealed journals is not measured (SHOGEN-ASN-PARTIELLE-S2-1, annex B.49).

Glossary (French → English)

A. Passages kept in French (printed outputs, verdicts and sealed labels)

French, as printed	English gloss
NE REJETTE PAS	does not reject
« R1 discrimine » = FAUX	“R1 discriminates” = false
« le modèle binomial de doc 10 §5.1, à fenêtres indépendantes, est rejeté ; la cause n'est pas identifiée entre co-défaillance des sources et dépendance sérielle des fenêtres »	the binomial model of doc 10 §5.1, with independent windows, is rejected; the cause is not identified between co-failure of the sources and serial dependence of the windows
« R1 rejette (s) »	R1 rejects (s)
hors décision	outside the decision
VRAI	true
« plage incluse »	range included
[SENSIBILITÉ]	SENSITIVITY
« rendu »	rendering
REJETTE	rejects
« BTC/USD-stable, devise marquée par flux »	BTC/USD-stable, currency marked per feed
« pool configuré = 12 flux »	configured pool = 12 feeds
« k nominal = 10 (hôtes distincts du pool) »	nominal k = 10 (distinct hosts of the pool)
« ok = 0 / 24585 lectures, n = 24585 »	ok = 0 / 24585 readings, n = 24585
« 10 sources / 11 flux »	10 sources / 11 feeds
« ok = 0 / 11397 lectures, n = 11397 »	ok = 0 / 11397 readings, n = 11397
« fenêtres distinctes (window_close) calme 0, stress 0 ; asn_attribution 0 ; clock_check 0 »	distinct windows (window_close) calm 0, stress 0; asn_attribution 0; clock_check 0
« causalité non établie »	causality not established
« kind=weekend_utc stress = samedi+dimanche UTC »	kind=weekend_utc stress = Saturday+Sunday UTC
« harnais vivant »	harness alive
« le résidu est à son MAXIMUM, publié »	the residual is at its MAXIMUM, published
« le z confirmatoire inclut les modes communs de l'observateur (hôte, DNS, réseau) »	the confirmatory z includes the common modes of the observer (host, DNS, network)
« n'est pas rejeté »	is not rejected
« aucune strate ne rejette ; strate(s) testée(s) : calme, stress »	no stratum rejects; stratum(s) tested: calm, stress
« nonÉv »	not evaluable
(panne / stale / horsE / nonÉv)	outage / stale / out-of-envelope / not evaluable
« côté livraison »	delivery side
« comparaison hétérogène déclarée »	heterogeneous comparison declared
« historique insuffisant »	insufficient history
« co-défaillance observée non expliquée par les axes R2 »	co-failure observed, not explained by the R2 axes
ÉTEINT	off
« aucun cluster à ≥ 2 flux (1) → tout reste au ∅ par paire de flux (bloc 4) »	no cluster with ≥ 2 feeds (1) → everything stays at the ∅ per pair of feeds (block 4)
« chemin de LECTURE ≠ amont (§3.2) : l'ASN mesuré est celui du fournisseur RPC (ethereum-rpc.publicnode.com), une infrastructure de lecture, pas l'amont du feed »	READING path ≠ upstream (§3.2): the measured ASN is that of the RPC provider (ethereum-rpc.publicnode.com), a reading infrastructure, not the upstream of the feed

French, as printed	English gloss
« bornes à $\hat{P}_{\text{more}}$ fixé, non extérieures ; verdict non identifié sous censure arbitraire des fenêtres sautées »	bounds at fixed $\hat{P}_{\text{more}}$, non-outer; verdict not identified under arbitrary censoring of the skipped windows
« non extérieures »	non-outer
« run maximal $\geq \ell$: $\hat{\sigma}^2_{\text{bloc},s}$ biaisé vers le bas ; SHOGEN-DEP-FENETRES-2 prioritaire avant G10 »	maximal run $\geq \ell$: $\hat{\sigma}^2_{\text{bloc},s}$ biased downward; SHOGEN-DEP-FENETRES-2 a priority before G10
« [inféré : dérivation de l'AVIS-advisor-defi Q1 (iv)] »	[inferred: derivation from the AVIS-advisor-defi Q1 (iv)]
« seconde coupe »	second cut
NON ÉVALUABLE : strate non testée, hors décision (§1 bis.1 pt 2)	not evaluable: stratum not tested, outside the decision (§1 bis.1 pt 2)
« $k_{\text{eff}} \leq 10$ (BORNE SUPÉRIEURE) »	$k_{\text{eff}} \leq 10$ (UPPER BOUND)
« ni levé ni éteint »	neither raised nor off
« aucune strate testée »	no stratum tested
« strate poolée hors des entrées »	pooled stratum outside the inputs
« variante “exclue” = principale (blocs 1-6) ; variante “incluse” = sensibilité — biaisée vers le haut par construction ; documente l'exclusion D5 ; pas un estimateur alternatif »	variant ‘excluded’ = main (blocks 1-6); variant ‘included’ = sensitivity — biased upward by construction; documents the D5 exclusion; not an alternative estimator
« motif de l'exclusion (harnais dégradé, [...]) : causalité non établie »	reason for the exclusion (degraded harness, [...]): causality not established
sensibilité « plage incluse » de la liste fermée (D2 pt 7), hors décision, biaisée vers le haut par construction	“range included” sensitivity of the closed list (D2 pt 7), outside the decision, biased upward by construction
« gardes levées »	guards cleared
« conforme »	conformant
« vivant »	alive
« à son MAXIMUM, publié »	at its MAXIMUM, published
« aucun cluster à ≥ 2 flux (1) »	no cluster with ≥ 2 feeds (1)
« run_params divergents »	diverging run_params
« ajoutée après le pré-enregistrement, hors décision »	added after the pre-registration, outside the decision
« ajoutée après le pré-enregistrement, hors décision ; ne change pas le verdict de la règle scellée (« R1 discrimine » = FAUX, docs/11 §3) »	added after the pre-registration, outside the decision; does not change the verdict of the sealed rule (“R1 discriminates” = false, docs/11 §3)
« go exécution »	go execution

B. Terms

French	English	Note
k_{eff} ; k nominal ; k nominal_s	k_{eff} ; nominal k ; nominal k_s	k_{eff} : number of classes of the measured R2 partition; never an independence count
enveloppe ; hors-enveloppe	envelope ; out-of-envelope	
fenêtre ; fenêtres sautées	window ; skipped windows	one minute of the UTC grid
strate ; strate calme ; strate de stress ; strate poolée	stratum (pl. strata) ; calm stratum ; stress stratum ; pooled stratum	printed stratum names: calme, stress
flux	feed	a reading of the BTC/USD (or BTC/USDT) price served by a source
source ; place (d'échange) ; agrégateur ; oracle	source ; exchange ; aggregator ; oracle	
hôte ; hôte de collecte	host ; collection host	
panne	outage	status panne_transport kept as printed
écart (doc 10 §5.2) ; en écart ; co-écart ; taux d'écart	anomaly ; anomalous ; co-anomaly ; anomaly rate	a feed is anomalous in a window if in outage, stale or out of envelope
écart (sens courant) ; écart déclaré (à un texte) ; écart (contrôle d'horloge)	difference, gap ; declared departure ; offset	offset: clock-check sense (§9.2 pt 17), as in the median offset of §11.1
déviations (déclarées)	(declared) deviation	departure from the pre-registered plan
co-défaillance	co-failure	
discordance	discordance	rule case: $z_s \geq 2.33$ and published $z_{\text{bloc},s} < 2.33$
sceau ; sceller ; scellé ; scellement	seal ; to seal ; sealed ; sealing	
paquet (de pré-enregistrement) ; paquet scellé	(pre-registration) package ; sealed package	
rendu (nom)	rendering	an output file of the single execution (SUITE, J14p, J14s, J28, RT, RAW)
exécution unique	single execution	
sortie ; sortir 0	output ; exit 0	
journal ; journaux scellés ; journal brut	journal ; sealed journals ; raw journal	JOURNAL: the project journal
relevé	probe record	diagnostic record of the harness (asn_attribution, clock_check)
enregistrement (d'oracle)	(oracle) record	
lecture	reading	a data reading; also an interpretation; also an access listed in annex D.1
rédacteur (frais) ; auteur ; validateur ; relecture	(fresh) drafter ; author ; validator ; review	
valider	accept (at review)	the assurance qualifier listed in doc 09 is not used (gate S-G4)
orchestrateur ; investisseur ; sous-agent	orchestrator ; investor ; sub-agent	
feu vert ; brouillon	go-ahead ; draft	

French	English	Note
dépôt (privé) ; versé	(private) repository ; filed	
harnais ; démarrage	harness ; start	
garde ; déclencheur	guard ; trigger	
drapeau ; levé ; éteint	flag ; raised ; off	printed state kept: ÉTEINT, ETEINT
seuil ; puissance	threshold ; power	
erreur-type ; erreur-type par blocs ; plancher d'erreur-type par blocs	standard error ; block standard error ; block standard-error floor	
excès minimal détectable	minimum detectable excess	EMD_s
statistique confirmatoire	confirmatory statistic	
censure ; bornes (non) extérieures	censoring ; (non-)outer bounds	
plage (D5) ; segment ; coupe	(D5) range ; segment ; cut	
variante exclue ; variante incluse	excluded variant ; included variant	
sensibilité ; descriptif	sensitivity ; descriptive	
bloc ; bloc machine	block ; machine block	
jeton ; manifeste ; horodatage ; autorité d'horodatage	token ; manifest ; timestamp ; timestamping authority	
délai de rétractation ; tête gardée ; voie (a)	withdrawal period ; guarded head ; route (a)	
recalcul tiers ; chemin de recalcul ; replay	third-party recomputation ; recomputation path ; replay	
commit d'analyse ; commit de collecte	analysis commit ; collection commit	
amont ; amont déclaré	upstream ; declared upstream	
résidu ; chemin de lecture ; chemin de mesure	residual ; reading path ; measurement path	
mono-vantage ; sonde multi-résolveurs	single-vantage ; multi-resolver probe	
constat ; [constat du rapport] ; inféré	finding ; [report finding] ; inferred	[calc]: drafter's computation
recopié ; imprimé ; étiquette	copied (verbatim outputs) or quoted (translated quotations) ; printed ; label	a translated quotation is not a verbatim copy; the original prevails
limite ; lot ; annexe	limitation ; work package ; annex	
pt, pts ; n° ; l.	pt, pts ; No. ; l.	point(s) ; number ; line
exposition ; poste local ; session cloud	exposure ; local workstation ; cloud session	
fonction de difficulté	difficulty function	Littlewood and Miller (L&M)
témoignage attesté	attested statement	doc 09
J14 principal ; J14 second	main J14 ; second J14	run names kept: j14-principal, j14-second

Source file sha256: 60f7dc7a7b6784a2d528e5d5f2ffe1798bef20b20c12156be688e92a7b2021d6